Physical Biology of the Cell

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How to read these notes

The main text follows a conceptual progression from quantities and rate models to probability, diffusion, statistical mechanics, and useful representations. The appendices provide mathematical tools, detailed derivations, reading lists, the original daily sequence, and notation.

Mathematical methods reference

Appendix A is a working reference for readers who have not previously encountered all the mathematical methods used in these notes. It focuses on how to read the notation, what the mathematical objects represent, how to calculate with them, and when each method is useful. The appendix is not intended to be read linearly before the main text. When a mathematical step becomes unfamiliar, consult the corresponding topic below.

	Concept	**Section to consult**
1.	Reading notation, exponentials, logarithms, units, and scales	Sections A.1 and 1.1
2.	Derivatives, integrals, partial derivatives, and conservation	Section A.2
3.	Taylor expansions and interpolation between known points	Sections A.3-A.4
4.	ODEs, master equations, PDEs, and time evolution	Sections A.5-A.7
5.	Vectors, matrices, eigenvectors, and eigenfunctions	Sections A.8-A.9
6.	Fourier series, spatial modes, and delta distributions	Section A.10
7.	Probability, expectation, variance, and waiting times	Sections A.11-A.12
8.	Counting states, Boltzmann weights, free energy, and entropy	Sections A.13-A.15
9.	Least squares, covariance, correlation, and PCA	Sections A.16-A.17

Acknowledgements

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1 What is life? Thinking quantitatively about living systems

The opening questions concern what makes a system living and what a physical explanation can reveal. Order-of-magnitude thinking then turns broad questions into quantities that can be estimated and compared. A useful thought experiment is to ask which properties of life on Earth might be universal and which might simply be particular implementations that arose on Earth. If life existed elsewhere, some of its organizing principles might be recognizable even if its molecular details were completely different.

Potentially general feature	**Earth-specific implementation that need not be universal**
A self-replicating blueprint	DNA and its particular molecular structure
Organization into self-replicating units	Cells as we know them
Metabolism and energy transduction	Glycolysis, ATP, and particular terrestrial metabolic pathways
Adaptation and/or evolution	Particular terrestrial evolutionary mechanisms and molecular machinery
Birth and death of individual systems	Particular cellular life cycles
A chemical basis for structure and function	CHNOPS chemistry
Error correction and maintenance of information	Particular DNA-repair and proofreading mechanisms

Table 1: A thought experiment about which features of terrestrial life might reflect general principles of living systems and which may be contingent molecular implementations.

The phrase "the secret of life" became associated with the discovery of the structure of DNA through a famous anecdote. According to James Watson's later account, after he and Francis Crick arrived at their double-helix model in 1953, Crick went into the Eagle pub in Cambridge and announced that they had "found the secret of life" [1]. Their model suggested how genetic information could be copied through complementary base pairing [2, 3].

About a decade later, Jacques Monod reportedly used the same language when he said to his colleague Agnes Ullmann, "I think I have discovered the second secret of life": allostery [4]. Allostery is functional coupling between distinct sites on a protein, such that binding at one site changes binding or activity at another [5]. We revisit allostery in section 5.6.

1.1 The scale of things: order-of-magnitude and Fermi thinking

Order-of-magnitude estimates and Fermi problems use a small number of known scales and physical constraints to obtain approximate answers to biological questions [6]. When a quantity is known only to lie somewhere between two scales, $x_{\min}$ and $x_{\max}$, a useful order-of-magnitude estimate is the geometric mean,

$$x_{\text{est}} \sim \sqrt{x_{\min} x_{\max}}.$$

This is especially natural when the uncertainty spans orders of magnitude. On a logarithmic scale,

$$\log x_{\text{est}} = \frac{1}{2}\left(\log x_{\min} + \log x_{\max}\right),$$

so the estimate lies halfway between the two bounds in log-space.

Units and dimensional consistency

Quantities that are added or equated must have the same dimensions. For example, in the relation $\ell = vt$, the distance ℓ has dimensions of length, so the velocity v must have dimensions of length divided by time. Arguments of exponential, logarithmic, and trigonometric functions must be dimensionless. Thus, in an exponential decay $e^{-t/\tau}$, the characteristic timescale τ must have the same units as t, making the ratio t/τ dimensionless.

Dimensional reasoning can also identify a characteristic scale before a detailed calculation. The diffusion constant has dimensions

$$[D] = \frac{\text{length}^2}{\text{time}}.$$

The characteristic time constructed from a length ℓ and a diffusion constant D is therefore $t_D \sim \ell^2/D$. For $\ell = 10\,\mu\text{m}$ and $D = 1\,\mu\text{m}^2\text{s}^{-1}$, this gives $t_D \sim 100\,\text{s}$. The numerical prefactor depends on the dimension, geometry, and precise meaning of "diffuse across."

A practical Fermi-estimation protocol

1. State the quantity and its desired units.
2. Decompose it into factors that can be estimated.
3. Assign a plausible scale to each factor.
4. Multiply, keeping powers of ten and units explicit.
5. Check the result against obvious lower and upper bounds.
6. Report the assumptions and only the precision justified by the estimate.

As an illustrative estimate, take deliberately rounded bacterial-scale values: a ribosome adds roughly 20 amino acids per second, a typical protein contains roughly 300 amino acids, approximately 10 ribosomes translate one mRNA simultaneously, and the mRNA has an effective lifetime of about five minutes. These values are representative rather than universal. One ribosome completes a protein in approximately $300/20 \sim 15\,\text{s}$, so ten ribosomes produce of order $10/15 \sim 0.7$ proteins per second. Over this five-minute timescale, this gives roughly $0.7 \times 300 \sim 200$ proteins. The point is the scale and the assumptions, not a falsely precise value. Further biological examples and experimentally measured scales are collected in the BioNumbers database [7], and are also available at book.bionumbers.org.

1.2 Case study: inferring a lipid bilayer from areas

Gorter and Grendel used red blood cells to infer that cellular membranes consist of a lipid bilayer [8].

The basic logic was:

1. Take a known number of red blood cells.
2. Estimate the total surface area of those cells.
3. Extract the lipids from the cells.
4. Spread the extracted lipids into a monolayer.
5. Measure the area occupied by this lipid monolayer.

The lipid monolayer occupied approximately twice the area of the cell surfaces. This suggests that the membrane contains two layers of lipid:

$$A_{\text{lipid}} \approx 2A_{\text{cell}}.$$

Hence the membrane is approximately a lipid bilayer.

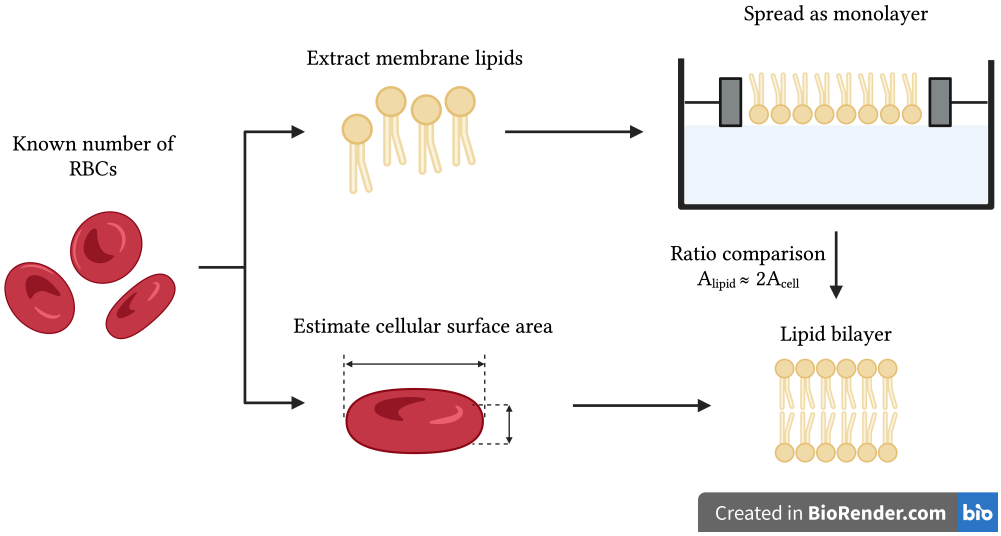


Figure 1: The Gorter-Grendel experiment [8]. Lipids extracted from a known number of red blood cells were spread into a monolayer and compared with the estimated total surface area of the cells. The lipid monolayer occupied approximately twice the cell-surface area, $A_{\text{lipid}} \approx 2A_{\text{cell}}$, providing evidence that the cell membrane consists of a lipid bilayer. Created with BioRender.com.

2 Modeling biological processes: variables, rates, and time evolution

A model starts with a question and a choice of variables. For growth, a single mass variable may suffice; for binding, concentrations of free and bound molecules are useful. The aim is to choose a description that retains the processes needed to answer the question. The broader problem of finding useful representations is revisited in section 6.

2.1 The theory of the mean

A useful way of constructing a dynamical model is to keep track of the mean value of one or more variables of interest.

The basic protocol is:

1. Identify the variables of interest.
2. Enumerate all the ways in which those variables can increase or decrease, and assign rates to those processes.
3. Write an update rule describing the change during a small time interval Δt.
4. Take the continuum limit and integrate the resulting differential equation to obtain the time evolution.

2.2 Autocatalysis and exponential cellular growth

A useful coarse-grained picture of the cell is as a system whose dry mass is dominated by proteins and whose growth is therefore closely related to protein production [9]. In *E. coli*, measured protein-to-mRNA abundance ratios span approximately 10^2 - 10^4 across genes, making

$$\frac{\text{number of protein molecules}}{\text{number of mRNA molecules}} \sim 10^3$$

a useful order-of-magnitude scale [10].

Ribosomes synthesize proteins, including the ribosomal proteins required to construct new ribosomes. Thus the translational machinery participates in its own production. At a coarse-grained level, this creates an *autocatalytic* process: the machinery responsible for growth produces more of the machinery responsible for growth [11]. This provides an intuitive route to exponential cellular growth.

Simple growth law

Let $M(t)$ denote the cell mass at time t.

Assume that the rate at which new mass is produced is proportional to the mass already present. Over a short time interval Δt,

$$M(t + \Delta t) = M(t) + kM(t)\Delta t,$$

where k is the fractional growth rate. Rearranging,

$$\frac{M(t + \Delta t) - M(t)}{\Delta t} = kM(t).$$

Taking the limit $\Delta t \to 0$ gives

$$\frac{\mathrm{d}M}{\mathrm{d}t} = kM.$$

The solution is

$$M(t) = M(0)e^{kt}.$$

Thus a constant fractional growth rate produces exponential growth.

This model assumes that

- the cell can be described by a single coarse-grained variable $M(t)$;
- the amount of active protein-production machinery is approximately proportional to cell mass;
- the protein-production rate per unit mass is approximately constant;
- nutrients and other resources are not limiting over the time interval considered;
- degradation and other losses are either negligible or already incorporated into the effective rate constant k;
- fluctuations in individual molecular events can be neglected, so that a deterministic continuum description is appropriate.

2.3 Biological turnover

Biological systems continually replace components such as red blood cells, photoreceptor components, proteins, RNA molecules, and cells in renewing tissues [12]. Human red blood cells provide a classic example: isotope-labeling experiments established that they have a finite lifespan and are continuously replaced [13]. A simple birth-death model illustrates how turnover can produce a steady state even while individual components are continuously produced and destroyed. The same general modeling framework applies: identify the states of interest, enumerate the processes that move material between those states, assign rates, and construct the corresponding dynamical equations.

2.4 Example: phytoplankton resource competition

A biological example of constructing dynamical rules from rates is the uptake of limiting resources by phytoplankton. David Tilman studied competition between freshwater phytoplankton species under different supplies of limiting resources. The experiments compared predictions from quantitative resource-competition models with the observed abundances of competing algal species [14].

2.5 Case study: mRNA production, degradation, and steady state

Let $m(t)$ denote the mean number of mRNA molecules per cell at time t.

Suppose that

- mRNA molecules are produced at a constant rate r;
- each existing mRNA has a constant probability per unit time γ of being degraded.

For example, consider a production rate

$$r \sim 1 \text{ mRNA per few minutes},$$

while the characteristic mRNA lifetime is

$$\tau = \frac{1}{\gamma}.$$

Over a small time interval Δt,

$$\begin{aligned} m(t + \Delta t) &= m(t) + \text{mRNAs produced} - \text{mRNAs degraded} \\ &= m(t) + r\Delta t - \gamma m(t)\Delta t. \end{aligned}$$

Therefore,

$$\frac{m(t + \Delta t) - m(t)}{\Delta t} = r - \gamma m(t).$$

Taking $\Delta t \to 0$,

$$\frac{\mathrm{d}m}{\mathrm{d}t} = r - \gamma m. \tag{1}$$

Dimensional consistency determines the units of the parameters. The left-hand side has units of molecules per time. Therefore r must have units of molecules per time and γ must have units of inverse time, so that γm also has units of molecules per time. The product γt is dimensionless, as required when it appears in the exponential $e^{-\gamma t}$.

At steady state,

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 0,$$

so

$$m_{\text{ss}} = \frac{r}{\gamma}.$$

The full solution is

$$m(t) = \frac{r}{\gamma} + \left(m(0) - \frac{r}{\gamma} \right) e^{-\gamma t}.$$

Thus the system approaches its steady-state mRNA number on the characteristic timescale

$$\tau = \frac{1}{\gamma}.$$

The deterministic model assumes that

- transcription occurs at a constant average rate r;
- degradation is first-order: each mRNA has the same constant degradation probability per unit time;
- mRNA molecules act independently;
- r and γ do not vary with time;
- discrete molecular fluctuations are ignored.

At the level of individual molecules, transcription and degradation are stochastic processes. Single-cell measurements directly reveal fluctuations in mRNA copy number and burst-like transcription [15, 10]. Eqn. (1) describes the mean behavior.

2.6 Case study: ligand-receptor binding

Consider a ligand L that can bind reversibly to a receptor R to form a ligand-receptor complex LR.

The variables of interest are therefore the concentrations

$$[R], \qquad [L], \qquad [LR].$$

Here $[R]$ denotes the concentration of free receptors, $[L]$ the concentration of free ligands, and $[LR]$ the concentration of bound ligand-receptor complexes.

Step 1: identify the variables

A schematic representation of the three states is

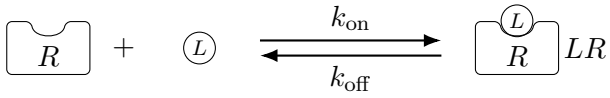


Figure 2: Reversible binding of a ligand L to a receptor R to form the complex LR. The same ligand and receptor symbols are used in the unbound and bound states.

Step 2: enumerate the processes and their rates

The reaction is

$$R + L \underset{k_{\text{off}}}{\overset{k_{\text{on}}}{\rightleftharpoons}} LR.$$

The forward reaction is ligand binding. According to the law of mass action, its rate is proportional to both the free ligand concentration and the free receptor concentration,

$$\text{binding rate} = k_{\text{on}}[L][R].$$

The reverse reaction is ligand unbinding. Its rate is proportional to the concentration of bound complexes,

$$\text{unbinding rate} = k_{\text{off}}[LR].$$

The association rate constant k_{on} has units of concentration^{-1} time^{-1}. The dissociation rate constant k_{off} has units of time^{-1}.

Step 3: write the update rule

During a short time interval Δt, the concentration of bound complexes changes according to

$$[LR](t + \Delta t) = [LR](t) + k_{\text{on}}\Delta t[L][R] - k_{\text{off}}\Delta t[LR].$$

The first additional term represents newly formed complexes. The second represents complexes lost through dissociation. Subtracting $[LR](t)$ from both sides and dividing by Δt gives

$$\frac{[LR](t + \Delta t) - [LR](t)}{\Delta t} = k_{\text{on}}[L][R] - k_{\text{off}}[LR].$$

Taking the limit $\Delta t \to 0$ gives

$$\boxed{\frac{\mathrm{d}[LR]}{\mathrm{d}t} = k_{\text{on}}[L][R] - k_{\text{off}}[LR].}$$

Steady state

At steady state, the mean concentration of bound complexes is no longer changing,

$$\frac{\mathrm{d}[LR]}{\mathrm{d}t} = 0.$$

Therefore,

$$k_{\mathrm{on}}[L][R] - k_{\mathrm{off}}[LR] = 0.$$

Hence

$$k_{\mathrm{on}}[L][R] = k_{\mathrm{off}}[LR].$$

Rearranging,

$$\frac{[L][R]}{[LR]} = \frac{k_{\mathrm{off}}}{k_{\mathrm{on}}}.$$

We define the dissociation constant

$$\boxed{K_d = \frac{k_{\mathrm{off}}}{k_{\mathrm{on}}}.}$$

Therefore,

$$K_d = \frac{[L][R]}{[LR]}.$$

Equivalently,

$$\frac{[LR]}{[R]} = \frac{[L]}{K_d}.$$

Probability that a receptor is bound

The total receptor concentration is the sum of free and bound receptors,

$$[R]_{\mathrm{tot}} = [R] + [LR].$$

The fraction of receptors that are bound is therefore

$$P_{\mathrm{bound}} = \frac{[LR]}{[R] + [LR]}.$$

Using

$$\frac{[LR]}{[R]} = \frac{[L]}{K_d},$$

we obtain

$$P_{\mathrm{bound}} = \frac{[L]/K_d}{1 + [L]/K_d}.$$

Thus

$$\boxed{P_{\mathrm{bound}}([L]) = \frac{[L]}{K_d + [L]}.} \tag{2}$$

This expression has several useful limits.

When there is no ligand,

$$[L] = 0 \quad \Rightarrow \quad P_{\mathrm{bound}} = 0.$$

When the ligand concentration is very large compared with K_d,

$$[L] \gg K_d \quad \Rightarrow \quad P_{\text{bound}} \to 1.$$

When

$$[L] = K_d,$$

we have

$$P_{\text{bound}} = \frac{1}{2}.$$

Thus K_d is the ligand concentration at which half of the receptors are occupied in this binding model. A smaller K_d corresponds to stronger binding, since less ligand is required to reach half occupancy.

Step 4: time evolution

To solve explicitly for the time evolution, suppose that the free ligand concentration $[L]$ remains approximately constant. The total receptor concentration is conserved,

$$[R]_{\text{tot}} = [R] + [LR].$$

Therefore,

$$[R] = [R]_{\text{tot}} - [LR].$$

Substituting this into the dynamical equation gives

$$\frac{\mathrm{d}[LR]}{\mathrm{d}t} = k_{\text{on}}[L]\left([R]_{\text{tot}} - [LR]\right) - k_{\text{off}}[LR].$$

Collecting terms gives

$$\frac{\mathrm{d}[LR]}{\mathrm{d}t} = k_{\text{on}}[L][R]_{\text{tot}} - \left(k_{\text{on}}[L] + k_{\text{off}}\right)[LR].$$

The equilibrium concentration is

$$[LR]_{\text{eq}} = [R]_{\text{tot}} \frac{[L]}{K_d + [L]}.$$

The full time-dependent solution is

$$[LR](t) = [LR]_{\text{eq}} + \left([LR](0) - [LR]_{\text{eq}}\right) e^{-(k_{\text{on}}[L] + k_{\text{off}})t}.$$

The system therefore relaxes exponentially toward equilibrium with characteristic timescale

$$\tau = \frac{1}{k_{\text{on}}[L] + k_{\text{off}}}.$$

The model assumes that:

- the system is well mixed, so spatial variation in ligand and receptor concentrations can be ignored;
- each receptor has a single independent ligand-binding site;
- binding follows the law of mass action;

- the rate constants k_{on} and k_{off} are constant;
- receptors and ligands are chemically identical within their respective populations;
- there is no cooperativity between receptors;
- there are no additional receptor states or competing ligands;
- for the explicit time-dependent solution above, the free ligand concentration $[L]$ is treated as constant.

The same steady occupancy will be recovered by counting microscopic states in section 5.3. The rate description additionally gives the relaxation time; an equilibrium occupancy alone does not determine both rate constants.

3 Statistics: from means to distributions and null models

A mean can conceal variation between cells, molecules, or populations. Probability distributions specify that variation; a master equation specifies how the distribution changes. The mRNA example links this chapter directly to the deterministic model in section 2.5.

3.1 Probability and quantitative null hypotheses

A useful null hypothesis is the simplest random process that could plausibly generate the observations. The purpose of the null hypothesis is to ask what behavior we should expect even without invoking an additional mechanism. Probability distributions provide mathematical descriptions of the fluctuations expected under different random processes.

For a discrete random variable, probabilities sum to one and expectations are probability-weighted sums. For a continuous variable, a probability density integrates to one and expectations are integrals. Appendix A.11 summarizes normalization, expectation, and variance.

3.2 Recurring probability distributions

Different stochastic processes naturally give rise to different probability distributions. For a discrete variable, probabilities are specified by a probability mass function (PMF). For a continuous variable, they are specified by a probability density function (PDF), whose integral over an interval gives the probability of that interval.

Some recurring distributions in physical biology are (see Table 2):

1. binomial
2. Poisson
3. Gaussian
4. exponential
5. geometric

The binomial, Poisson, and Gaussian distributions recur throughout the course examples. The exponential distribution describes an important new class of problems: waiting times.

3.2.1 Waiting times and molecular binding

The exponential distribution describes the waiting time until an event occurs when that event happens with a constant rate. Let the event occur at rate λ. The probability density for the waiting time t is

$$\boxed{p(t) = \lambda e^{-\lambda t}, \qquad t \geq 0.}$$

The mean waiting time is

$$\langle t \rangle = \frac{1}{\lambda},$$

and the variance is

$$\mathrm{Var}(t) = \frac{1}{\lambda^2}.$$

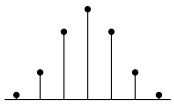
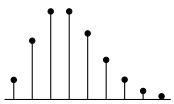
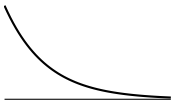
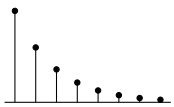
Distribution	PMF/PDF	Mean	Variance	Representative examples
Binomial 	$P(k) = \binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$	Receptor occupancy among n independent receptors; rightward steps in an n-step random walk; protein partitioning at cell division.
Poisson 	$P(k) = e^{-\lambda} \dfrac{\lambda^k}{k!}$	λ	λ	Mutations per genome; deaths from horse kicks per army corps; wartime bomb hits per district; photons detected during a fixed time interval.
Gaussian 	$p(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\dfrac{(x-\mu)^2}{2\sigma^2}\right]$	μ	σ^2	Errors in repeated measurements; displacement after many diffusive steps; polymer end-to-end displacement; variation in human height.
Exponential 	$p(t) = \lambda e^{-\lambda t}$	$\dfrac{1}{\lambda}$	$\dfrac{1}{\lambda^2}$	Waiting time until radioactive decay; time between constant-rate events; lifetime of a molecular state; waiting time until the next photon.
Geometric 	$P(n) = (1-p)^n p$	$\dfrac{1-p}{p}$	$\dfrac{1-p}{p^2}$	Failures before the first success; unsuccessful chromosome-capture attempts; tails before the first head; rejected proofreading attempts.

Table 2: Common probability distributions and representative examples. The geometric distribution is defined here as the number of failures before the first success.

Example: molecular binding times

Single-molecule fluorescence experiments can resolve individual association and dissociation events. For example, Wang et al. immobilized fluorescently labelled DNA molecules on a surface and observed the binding of fluorescently labelled λ repressor proteins [16]. When a protein was bound to a DNA molecule, the two fluorescence signals were observed at the same location. The resulting signal could be represented by a two-state time trace,

$$I(t) = \begin{cases} 1, & \text{protein bound,} \\ 0, & \text{protein unbound.} \end{cases}$$

A sequence of binding and unbinding events then appears as alternating intervals in which $I(t) = 1$ and $I(t) = 0$.

Bound-state lifetime

Suppose a bound protein dissociates with a constant rate k_{off}. Let $S(t)$ be the probability that a protein which was bound at time $t = 0$ is still bound after time t. During a small interval Δt, the probability of dissociation is

$$k_{\text{off}}\Delta t.$$

The probability of remaining bound is therefore

$$1 - k_{\text{off}}\Delta t.$$

The update rule is

$$S(t + \Delta t) = S(t)\left(1 - k_{\text{off}}\Delta t\right).$$

Taking the continuous-time limit gives

$$\frac{\mathrm{d}S}{\mathrm{d}t} = -k_{\text{off}}S.$$

With the initial condition $S(0) = 1$,

$$\boxed{S(t) = e^{-k_{\text{off}}t}.}$$

This is the probability that the protein remains bound for at least time t. The probability density for the actual bound lifetime t_{on} is obtained from

$$p(t_{\text{on}}) = -\frac{\mathrm{d}S}{\mathrm{d}t_{\text{on}}}.$$

Therefore,

$$\boxed{p(t_{\text{on}}) = k_{\text{off}}e^{-k_{\text{off}}t_{\text{on}}}.}$$

The mean bound lifetime is

$$\langle t_{\text{on}}\rangle = \frac{1}{k_{\text{off}}}.$$

Thus measuring the distribution of bound-state durations provides a way to estimate the dissociation rate k_{off}.

Waiting for binding

The unbound intervals can be treated similarly. If binding occurs with an effective constant rate λ_{on}, then

$$p(t_{\text{off}}) = \lambda_{\text{on}}e^{-\lambda_{\text{on}}t_{\text{off}}}.$$

If k_{on} is a bimolecular association rate constant and the protein concentration is $[P]$, then the effective rate is

$$\lambda_{\text{on}} = k_{\text{on}}[P].$$

The important distinction is therefore

$$\text{bound lifetime} \longleftrightarrow k_{\text{off}},$$

whereas

$$\text{unbound waiting time} \longleftrightarrow k_{\text{on}}[P].$$

3.3 The theory of the histogram

The theory of the mean follows only the average value of a variable. In many biological systems, however, the full distribution contains information that the mean alone cannot capture. Instead of asking only how the mean changes with time, we can ask how probability moves between different possible values or states.

Example 1: number of mRNA molecules in a fly embryo

Suppose we want to describe the spatial distribution of mRNA molecules along a fly embryo. We can discretize the embryo into small spatial boxes. Each box then contains some number of mRNA molecules. The collection of counts across boxes forms a spatial distribution rather than a single mean value.

Example 2: age distributions

Age is intrinsically continuous, but it can be discretized into bins such as years. For example,

$$[0,1), \qquad [1,2), \qquad [2,3), \qquad \ldots$$

We can then count how many individuals occupy each age bin. The resulting histogram is a discretized representation of the underlying age distribution.

3.3.1 Theory of the histogram protocol

A corresponding protocol for describing an entire probability distribution is:

1. Consider the population of interest as a probability distribution over a set of possible states or bins.
2. Enumerate all the ways in which individuals or probability can move between those bins.
3. Write an update rule describing the flow of probability into and out of each bin during a time step Δt.
4. Given an initial distribution $p(m, t_0)$, solve for the distribution $p(m, t)$ at later times.

Here m labels the state or bin of interest. For example, m could denote the number of mRNA molecules in a cell.

Instead of following only the mean $\langle m \rangle$, we now follow the full probability distribution

$$p(m, t).$$

The normalization condition is

$$\sum_m p(m, t) = 1.$$

This shift from the dynamics of a mean to the dynamics of a distribution is important whenever fluctuations are themselves biologically relevant.

3.3.2 Example: the dynamics of an mRNA histogram

Consider a population of identical cells. Instead of describing only the mean number of mRNA molecules per cell, we describe the entire distribution of mRNA copy numbers.

Let

$$p(m,t)$$

denote the probability that a cell contains exactly m mRNA molecules at time t. Equivalently, if we measure a large population of cells, $p(m,t)$ is the fraction of cells occupying the histogram bin corresponding to m mRNAs.

The distribution is normalized,

$$\sum_{m=0}^{\infty} p(m,t) = 1.$$

Step 1: identify the states

The possible states are the discrete mRNA copy numbers

$$m = 0, 1, 2, 3, \ldots$$

Each value of m corresponds to one bin of the histogram.

Step 2: enumerate the ways of moving between bins

Assume that mRNA is produced at a constant rate r. A transcription event increases the copy number by one,

$$m \longrightarrow m + 1.$$

Assume that each individual mRNA molecule is degraded independently at rate γ. If there are m molecules in the cell, the total degradation rate is therefore

$$\gamma m.$$

A degradation event decreases the copy number by one,

$$m \longrightarrow m - 1.$$

The transitions between neighboring histogram bins are therefore

$$\cdots \rightleftharpoons m-1 \rightleftharpoons m \rightleftharpoons m+1 \rightleftharpoons \cdots$$

with transcription moving probability to the right and degradation moving probability to the left.

Step 3: write the probability flow for one bin

To determine how $p(m,t)$ changes, enumerate all the ways probability can flow into and out of the bin m during a short time interval Δt. There are two ways to enter the state m. First, a cell containing $m-1$ molecules can produce one new mRNA. The corresponding probability flow is

$$r\Delta t\, p(m-1,t).$$

Second, a cell containing $m+1$ molecules can lose one mRNA through degradation. Since there are $m+1$ molecules available to degrade, the total degradation rate in this state is $\gamma(m+1)$. The corresponding probability flow is

$$\gamma(m+1)\Delta t\, p(m+1,t).$$

There are also two ways to leave the state m. A transcription event moves the system from m to $m+1$, giving an outgoing contribution

$$r\Delta t\, p(m,t).$$

A degradation event moves the system from m to $m-1$, giving an outgoing contribution

$$\gamma m\Delta t\, p(m,t).$$

Putting all of these contributions together gives the update rule

$$\begin{aligned} p(m,t+\Delta t) = p(m,t) &+ r\Delta t\, p(m-1,t) + \gamma(m+1)\Delta t\, p(m+1,t) \\ &- r\Delta t\, p(m,t) - \gamma m\Delta t\, p(m,t). \end{aligned}$$

Its structure is simply

$$\text{new probability} = \text{old probability} + \text{flowing in} - \text{flowing out}.$$

Step 4: take the continuous-time limit

Subtracting $p(m,t)$ from both sides and dividing by Δt gives

$$\begin{aligned} \frac{p(m,t+\Delta t)-p(m,t)}{\Delta t} = &\, rp(m-1,t) + \gamma(m+1)p(m+1,t) \\ &- rp(m,t) - \gamma mp(m,t). \end{aligned}$$

Taking the limit $\Delta t \to 0$ gives

$$\boxed{\frac{\partial p(m,t)}{\partial t} = rp(m-1,t) + \gamma(m+1)p(m+1,t) - rp(m,t) - \gamma mp(m,t)}. \tag{3}$$

This is the chemical master equation for constitutive mRNA production and degradation. It describes the dynamics of the entire histogram rather than only the mean mRNA number.

Reading the master equation as probability flow

Each term has a direct interpretation:

1. $rp(m-1,t)$: probability flowing into bin m from bin $m-1$ through transcription.
2. $\gamma(m+1)p(m+1,t)$: probability flowing into bin m from bin $m+1$ through degradation.
3. $-rp(m,t)$: probability leaving bin m through transcription.
4. $-\gamma mp(m,t)$: probability leaving bin m through degradation.

Therefore the equation is simply a bookkeeping equation for probability.

Boundary at zero mRNAs

The copy number cannot become negative. For $m = 0$, there is therefore no probability flow into the state from $m = -1$. The equation for the zero-mRNA bin is

$$\frac{\partial p(0,t)}{\partial t} = \gamma p(1,t) - r p(0,t).$$

The first term represents cells entering the $m = 0$ state when their last mRNA is degraded. The second term represents cells leaving the $m = 0$ state when a new mRNA is produced.

Steady-state histogram

At steady state, the probability of occupying each bin no longer changes with time,

$$\frac{\partial p(m,t)}{\partial t} = 0.$$

For this simple birth–death process, the steady-state distribution is a Poisson distribution,

$$p_{\mathrm{ss}}(m) = \frac{\lambda^m e^{-\lambda}}{m!},$$

where the parameter $\lambda = \frac{r}{\gamma}$. Therefore,

$$\boxed{p_{\mathrm{ss}}(m) = \frac{1}{m!}\left(\frac{r}{\gamma}\right)^m e^{-r/\gamma}}.$$

The mean of this distribution is

$$\langle m \rangle_{\mathrm{ss}} = \frac{r}{\gamma}.$$

Its variance is also

$$\mathrm{Var}(m)_{\mathrm{ss}} = \frac{r}{\gamma}.$$

Connection to the theory of the mean

The histogram description contains the mean description as one of its consequences. The mean mRNA number is defined from the probability distribution as

$$\langle m(t) \rangle = \sum_{m=0}^{\infty} m\, p(m,t).$$

Taking the mean of the master equation (3) gives

$$\boxed{\frac{\mathrm{d}\langle m \rangle}{\mathrm{d}t} = r - \gamma \langle m \rangle}.$$

This is exactly the deterministic mean equation (1). At steady state,

$$\langle m \rangle_{\mathrm{ss}} = \frac{r}{\gamma}.$$

The two approaches therefore answer different levels of the same question. The theory of the mean tells us how the average copy number evolves. The theory of the histogram tells us how the probability of every possible copy number evolves. The histogram description consequently contains information about fluctuations, variance, and the shape of the population distribution that is invisible if we follow only the mean.

The simple mRNA birth–death model assumes that:

- transcription occurs at a constant rate r independent of the current mRNA number;
- each mRNA molecule degrades independently at the same rate γ;
- transcription changes the copy number by exactly $+1$;
- degradation changes the copy number by exactly -1;
- the system is Markovian, so transition probabilities depend only on the current state;
- Δt is sufficiently small that the probability of two or more reaction events during the same time interval is of order Δt^2 and can be neglected;
- there is no transcriptional bursting, promoter switching, feedback, or other regulation;
- cells are treated as identical apart from the stochastic differences generated by transcription and degradation.

These assumptions define a deliberately simple null model rather than a universal description of transcription. Single-cell measurements show that real transcription can involve promoter switching and burst-like production, producing fluctuations not captured by the constitutive birth-death model [15].

A spatial profile across embryo positions and a copy-number histogram across cells use different bins. In the first case, the bin labels a position; in the second, it labels a molecular count. The update protocol must specify both the state variable and the population being represented.

3.4 Case study: the Buri experiment and genetic drift

Peter Buri's experiment provides a classic demonstration of genetic drift in small populations. Buri studied many independent populations of *Drosophila melanogaster*. In one experimental series, 107 replicate populations were followed for 19 generations. Each population was initiated with 16 heterozygous flies. At every subsequent generation, 8 males and 8 females were randomly selected from the offspring to found the next generation. The population size was therefore kept small, while the particular alleles passed to the next generation were determined partly by random sampling (Fig. 3) [17].

Allele frequency

Using the red, orange, and white notation from the lecture, suppose that red flies carry two copies of the red allele, orange flies are heterozygous, and white flies carry no copies of the red allele. If there are N_{red} red flies, N_{orange} orange flies, and N_{flies} flies in total, then the frequency of the red allele is

$$p_{\text{red}} = q(bw^{75}) = \frac{2N_{\text{red}} + N_{\text{orange}}}{2N_{\text{flies}}}.$$

The factor of two in the denominator appears because each diploid fly carries two copies of the locus. Initially, all 16 flies are heterozygous, so

$$p_{\text{red}} = \frac{16}{32} = \frac{1}{2}.$$

As an example, suppose a later generation contains 4 red, 2 white, and 10 orange flies. Then

$$p_{\text{red}} = \frac{2(4) + 10}{2(16)} = \frac{18}{32} = 0.5625.$$

Genetic drift

Even when the two alleles have the same fitness, random sampling means that their frequencies do not remain exactly constant from one generation to the next. One population may contain slightly more copies of the red allele, while another may contain slightly fewer. These small fluctuations accumulate over generations. Different replicate populations therefore wander away from the common initial allele frequency,

$$p_{\text{red}} = \frac{1}{2}.$$

Eventually, some populations reach

$$p_{\text{red}} = 1,$$

meaning that the red allele has become fixed. Others reach

$$p_{\text{red}} = 0,$$

meaning that the red allele has been lost and the alternative allele has become fixed. These states are absorbing: once an allele has disappeared from a closed population, random sampling alone cannot bring it back. Thus substantial differences between populations can arise from random sampling alone, without invoking a selective advantage. This random change in allele frequency is called *genetic drift*.

Each vial contains 16 diploid flies, so there are

$$2 \times 16 = 32$$

copies of the relevant locus.

Let n_R denote the number of red alleles in a vial. The possible values are

$$n_R = 0, 1, 2, \ldots, 32.$$

There are therefore 33 possible allele-count states.

The allele count n_R itself is a single discrete variable. However, a probability distribution over all possible allele counts can be represented by a vector with 33 components,

$$\mathbf{p} = \begin{pmatrix} p_0 \\ p_1 \\ \vdots \\ p_{32} \end{pmatrix}.$$

Here

$$p_i = P(n_R = i).$$

The vector therefore lives in a 33-dimensional vector space, although not every vector in that space is a valid probability distribution. The components must satisfy

$$p_i \geq 0$$

and

$$\sum_{i=0}^{32} p_i = 1.$$

Thus a stochastic evolutionary process can be viewed as the time evolution of a probability vector over the possible allele-count states.

The 33-component representation describes the distribution over allele counts. It is useful even when the distribution mean does not reveal how populations accumulate at the absorbing states. Section 6.1 extends the vector idea to other biological data.

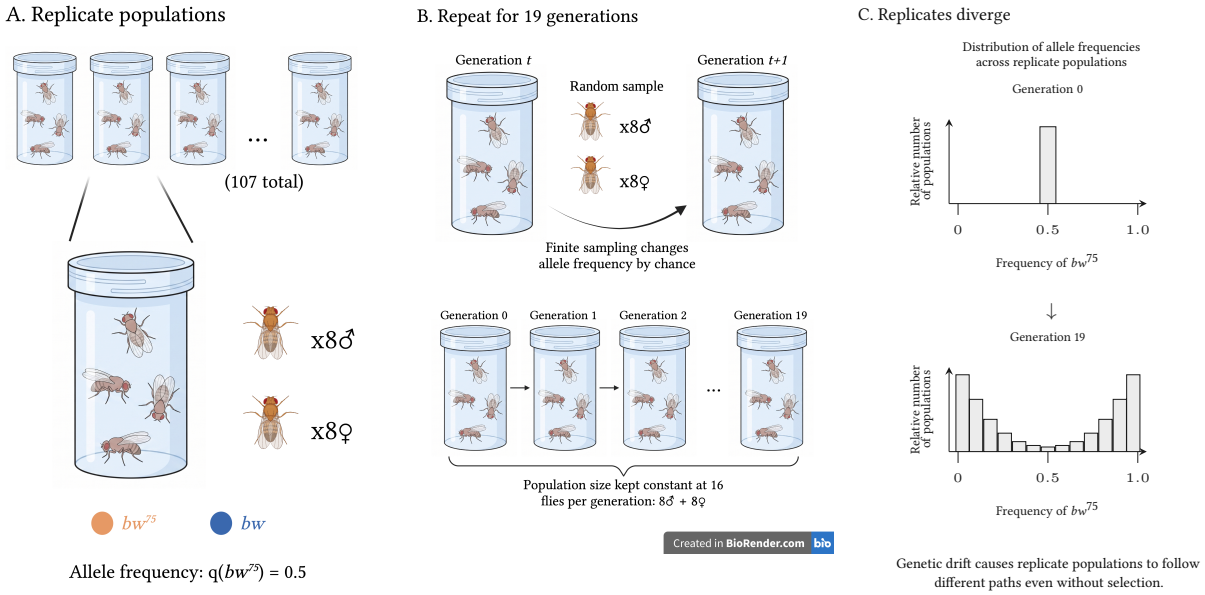


Figure 3: The Buri experiment and genetic drift. Replicate populations were initialized with 16 heterozygous flies, giving an initial frequency of $q(bw^{75}) = 0.5$. The bw^{75} allele frequency, denoted as $q(bw^{75})$ in (A), is written as p_{red} in the calculations above. Eight females and eight males were randomly selected to found each succeeding generation. Repeated sampling caused the allele-frequency distribution to spread toward loss and fixation. Panels A and B created with BioRender.com. Panel C is schematic and does not reproduce Buri's measured counts.

4 Random walks and diffusion

Spatial probability flow uses the same incoming-minus-outgoing bookkeeping as the mRNA histogram. A random walk makes the microscopic steps explicit; a lattice master equation evolves site probabilities; a continuum equation describes a smooth density.

4.1 Diffusion as a dynamical null hypothesis

Diffusion provides a useful dynamical null hypothesis for many biological processes. If there is no evidence for directed motion, active transport, interactions, external forces, or spatially varying dynamics, random motion provides the simplest starting model.

The same diffusive dynamics can be described at several different levels:

1. diffusion by coin flips or random walks;
2. diffusion by continuum theory;
3. diffusion by a master equation.

These are not fundamentally different phenomena. They are different mathematical descriptions of the same underlying stochastic motion.

4.2 Coin flips, spreading, and mean-square displacement

The most microscopic picture is a random walker making independent leftward or rightward jumps. In a discrete-time coin-flip model, the walker moves once during every step. In the continuous-time hopping model used below, jumps in either direction occur at rate k, so during a sufficiently short interval Δt the walker may jump left, jump right, or remain at its current position.

On a one-dimensional lattice with spacing a, each step changes the position by

$$x \to x + a \qquad \text{or} \qquad x \to x - a.$$

For an unbiased walk, the two directions are equally likely. Repeated independent random steps generate a probability distribution that spreads out with time. A characteristic signature of normal diffusion is

$$\langle x^2(t) \rangle \propto t.$$

In one dimension,

$$\langle x^2(t) \rangle = 2Dt,$$

where D is the diffusion constant [18].

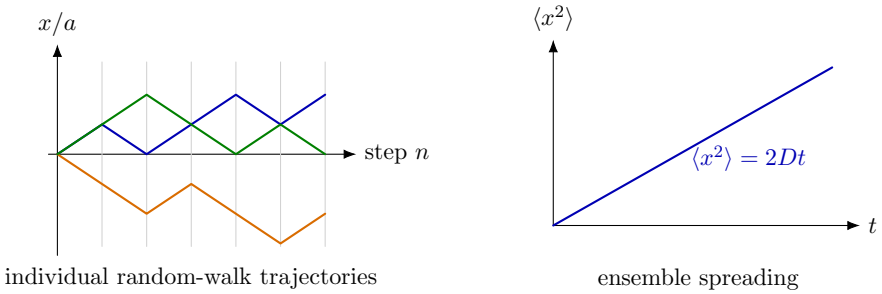


Figure 4: Independent left–right choices produce different trajectories, but the ensemble mean-square displacement grows linearly with time.

4.3 A one-dimensional symmetric random walk

Consider particles performing a random walk on a one-dimensional lattice. Define

- x: position,
- a: lattice spacing, i.e. the distance traveled in one jump,
- t: time,
- Δt: duration of one small time step,
- k: hopping rate in each direction,
- $P(x,t)$: probability of finding the particle at position x at time t.

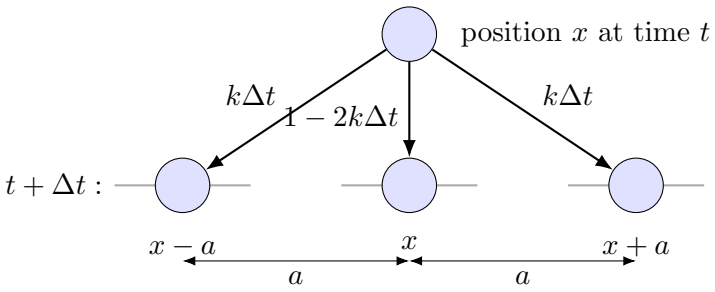


Figure 5: In one time step Δt, a walker at x moves by one lattice spacing a to either neighbor with probability $k\Delta t$, or remains at x with probability $1-2k\Delta t$.

During a short time interval Δt, the particle can

- jump from $x-a$ to x,
- jump from $x+a$ to x,
- jump away from x to either neighboring site,
- or remain at x.

The update rule is

$$\boxed{P(x,t+\Delta t) = P(x,t) + k\Delta t P(x-a,t) + k\Delta t P(x+a,t) - 2k\Delta t P(x,t)}$$

or equivalently,

$$P(x,t+\Delta t) = P(x,t) + k\Delta t\left[P(x-a,t) + P(x+a,t) - 2P(x,t)\right].$$

The quantity in square brackets is the discrete analogue of a second spatial derivative. The model assumes:

- motion occurs in one spatial dimension;
- space is homogeneous: the jump rate k is independent of x;
- the walk is unbiased: jumps to the left and right have the same rate;

- only nearest-neighbor jumps of length a are allowed;
- successive jumps are independent;
- the dynamics are Markovian: the next step depends only on the current state;
- Δt is sufficiently small that $2k\Delta t \leq 1$, so the jump probabilities remain valid;
- when Fourier modes $q_n = 2\pi n/L$ are used, the system is taken to be periodic with period L.

4.4 The lattice master equation

The random-walk update rule can also be viewed as a master equation for the probability of occupying each lattice site.

Let $P_i(t)$ denote the probability of occupying lattice site i. If particles hop from each site to either nearest neighbor with rate k, probability flows into site i from sites $i-1$ and $i+1$ and flows out of site i toward both neighbors. The continuous-time master equation is therefore

$$\boxed{\frac{\mathrm{d}P_i}{\mathrm{d}t} = kP_{i-1} + kP_{i+1} - 2kP_i.}$$

The first two terms are probability flowing into site i. The final term is probability flowing out of site i. Writing $x_i = ia$, we can identify

$$P_{i-1}(t) = P(x-a,t), \qquad P_i(t) = P(x,t), \qquad P_{i+1}(t) = P(x+a,t).$$

The master equation therefore becomes

$$\frac{\partial P}{\partial t} = k\left[P(x-a,t) + P(x+a,t) - 2P(x,t)\right].$$

Taylor-expanding in space once again gives

$$\frac{\partial P}{\partial t} = ka^2\frac{\partial^2 P}{\partial x^2} + \mathcal{O}(a^4).$$

Thus, in the continuum limit,

$$\boxed{\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}.}$$

The coin-flip picture, master-equation picture, and continuum diffusion equation are therefore three descriptions of the same process at different levels of resolution.

4.5 From the discrete update to the continuum equation

The same result can be obtained by taking the continuum limit of the random-walk update rule.

Start from

$$\frac{P(x,t+\Delta t) - P(x,t)}{\Delta t} = k\left[P(x-a,t) + P(x+a,t) - 2P(x,t)\right].$$

Taylor-expand the two spatially shifted terms:

$$P(x+a,t) = P(x,t) + a\frac{\partial P}{\partial x} + \frac{a^2}{2}\frac{\partial^2 P}{\partial x^2} + \mathcal{O}(a^3),$$
$$P(x-a,t) = P(x,t) - a\frac{\partial P}{\partial x} + \frac{a^2}{2}\frac{\partial^2 P}{\partial x^2} + \mathcal{O}(a^3).$$

Adding them,

$$P(x+a,t) + P(x-a,t) - 2P(x,t) = a^2\frac{\partial^2 P}{\partial x^2} + \mathcal{O}(a^4).$$

Therefore,

$$\frac{\partial P}{\partial t} = ka^2\frac{\partial^2 P}{\partial x^2}.$$

Since

$$D = ka^2,$$

we obtain the diffusion equation,

$$\boxed{\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}.} \tag{4}$$

The expanded derivation, the symmetry argument, and the continuum-limit assumptions are retained in appendix B.2. When comparing discrete and continuum normalization, a site probability is approximately the density multiplied by the lattice spacing: $P_i(t) \approx aP(x_i,t)$.

4.6 Fick's laws

Fick's first law states that the diffusive flux is proportional to the negative concentration gradient [19],

$$\boxed{J = -D\frac{\partial P}{\partial x}.}$$

The minus sign means that particles flow from regions of high concentration toward regions of low concentration. Conservation of particles gives the continuity equation,

$$\frac{\partial P}{\partial t} = -\frac{\partial J}{\partial x}.$$

Substituting Fick's first law,

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial x}\left(-D\frac{\partial P}{\partial x}\right).$$

For constant D, combining the continuity equation with Fick's first law gives Eqn. (4), also known as Fick's second law.

$$\boxed{\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}.}$$

This is Fick's second law, i.e. the diffusion equation.

4.7 Interpreting the continuum equation

Instead of following individual walkers, we can describe the probability density $P(x,t)$ continuously in space and time.

The continuum equation for diffusion in one dimension is

$$\boxed{\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}.}$$

The left-hand side describes how the probability distribution changes with time. The second spatial derivative measures the local curvature of the probability distribution. Diffusion therefore changes P wherever the distribution has spatial curvature. Peaks tend to flatten and valleys tend to fill in, causing the distribution to become progressively smoother.

4.7.1 The heat equation: the same mathematical structure

The heat equation has exactly the same mathematical form as the diffusion equation.

For temperature $T(x,t)$,

$$\boxed{\frac{\partial T}{\partial t} = \alpha\frac{\partial^2 T}{\partial x^2},}$$

where α is the thermal diffusivity. Thus

$$P(x,t) \longleftrightarrow T(x,t), \qquad D \longleftrightarrow \alpha.$$

The mathematics is the same even though the physical quantity being transported is different. For particle diffusion, spatial variations in particle concentration are smoothed out. For heat conduction, spatial variations in temperature are smoothed out. The diffusion equation is therefore not specific to particles. It describes a much broader class of processes in which some quantity spreads locally down spatial gradients.

4.8 Why Fourier modes are useful

The update rule contains translations of the probability distribution,

$$P(x-a,t), \qquad P(x+a,t).$$

Fourier modes are special because translating them does not change their shape: it only multiplies them by a number. The cleanest way to see this is with a complex exponential,

$$f_q(x) = e^{iqx}.$$

Translation gives

$$f_q(x+a) = e^{iq(x+a)} = e^{iqa}f_q(x),$$
$$f_q(x-a) = e^{-iqa}f_q(x).$$

Thus Fourier modes are eigenfunctions of translation.

Because the random-walk update rule is built entirely from translations, we should expect Fourier modes to diagonalize the dynamics. Cosines and sines are simply the real and imaginary combinations of these complex exponentials:

$$e^{iqx} = \cos(qx) + i\sin(qx).$$

This is the fundamental reason for trying sinusoidal modes.

4.8.1 Boundary conditions and allowed wave numbers

Suppose the system has length L and periodic boundary conditions,

$$P(x+L,t) = P(x,t).$$

A Fourier mode must then satisfy

$$e^{iq(x+L)} = e^{iqx}.$$

Therefore,

$$e^{iqL} = 1,$$

which requires

$$qL = 2\pi n, \qquad n \in \mathbb{Z}.$$

Hence the allowed wave numbers are

$$\boxed{q_n = \frac{2\pi n}{L}.}$$

4.8.2 Independent evolution of mode amplitudes

The diffusion equation contains a second spatial derivative,

$$\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}.$$

Fourier modes are eigenfunctions of this derivative operator because

$$\frac{\partial^2}{\partial x^2} e^{iqx} = -q^2 e^{iqx}.$$

Therefore the spatial derivative does not change the shape of a Fourier mode. It only multiplies that mode by $-q^2$.

For a Fourier component

$$P_q(x,t) = A_q(t)e^{iqx},$$

the diffusion PDE becomes

$$\frac{\mathrm{d}A_q}{\mathrm{d}t} = -Dq^2 A_q.$$

The solution is

$$A_q(t) = A_q(0)e^{-Dq^2 t}.$$

Hence

$$\boxed{P_q(x,t) = A_q(0)e^{-Dq^2 t}e^{iqx}.}$$

A PDE in space and time has been reduced to a simple ODE for the amplitude of each Fourier mode.

Yay, Fourier is great.

For a periodic density, the general solution is

$$P(x,t) = \frac{1}{L} + \sum_{n=1}^{\infty} \left[A_n \cos(q_n x) + B_n \sin(q_n x)\right] e^{-Dq_n^2 t}, \qquad q_n = \frac{2\pi n}{L}.$$

The coefficients specify the initial condition, and normalization fixes the constant mode to $1/L$. Cosines alone are sufficient for an even initial condition; a general periodic initial condition also needs sines. The discrete-mode calculation is in appendix B.1, and separation of variables is in appendix B.3. These are complementary ways of deriving the same long-wavelength mode decay.

4.9 Making a spike from cosines

A sharply localized distribution can be constructed from many Fourier modes. On a periodic domain of length L, the periodic Dirac delta has the Fourier representation

$$\boxed{\delta_L(x) = \frac{1}{L} \sum_{n=-\infty}^{\infty} e^{i2\pi nx/L}.}$$

Combining positive and negative frequencies gives

$$\boxed{\delta_L(x) = \frac{1}{L} \left[1 + 2 \sum_{n=1}^{\infty} \cos\left(\frac{2\pi nx}{L} \right) \right].}$$

Keeping only the first N modes gives

$$\delta_N(x) = \frac{1}{L} \left[1 + 2 \sum_{n=1}^{N} \cos\left(\frac{2\pi nx}{L} \right) \right].$$

Each cosine extends across the entire domain, so no single mode is localized near $x = 0$. When the modes are added, however, their peaks coincide at $x = 0$ while their positive and negative parts largely cancel elsewhere. Including more modes narrows the region in which they add constructively, producing an increasingly sharp peak around $x = 0$.

The finite Fourier sum is a representation of a delta distribution and may contain negative sidelobes. It need not itself be a nonnegative probability density. The delta limit is understood in the distributional sense.

4.10 Diffusion of an initially sharp distribution

Suppose the particle begins at $x = 0$, so that

$$P(x, 0) = \delta_L(x).$$

The Fourier representation is

$$P(x, 0) = \frac{1}{L} \left[1 + 2 \sum_{n=1}^{\infty} \cos(q_n x) \right].$$

Under diffusion, every mode acquires its own decay factor,

$$e^{-Dq_n^2 t}.$$

Therefore,

$$\boxed{P(x, t) = \frac{1}{L} \left[1 + 2 \sum_{n=1}^{\infty} e^{-Dq_n^2 t} \cos(q_n x) \right].}$$

At early times, many high-frequency modes are present and the distribution can be sharply peaked. At later times, the high-q modes have decayed because their decay rates

$$Dq_n^2$$

are large. Only the smooth, long-wavelength modes remain. Eventually all $n \neq 0$ modes disappear and

$$P(x, t) \longrightarrow \frac{1}{L}.$$

Thus diffusion can be understood as the progressive removal of high-frequency spatial structure.

5 Microscopic states to macroscopic behavior

The preceding chapters evolved variables or distributions in time. Here the central question is how available microscopic states and their weights determine macroscopic probabilities. Binding provides the connection between the rate-based and statistical-mechanical descriptions.

5.1 States, multiplicities, and statistical weights

A microstate specifies a microscopic arrangement; a macrostate groups arrangements according to an observable such as receptor occupancy. Multiplicity counts the microstates in a macrostate. At thermal equilibrium, a microstate with energy E_i has relative Boltzmann weight $e^{-\beta E_i}$, with $\beta = 1/(k_B T)$. Summing weights gives a partition function; dividing a state's weight by the sum normalizes its probability. If the W microstates in a macrostate share one energy E, its weight is $We^{-\beta E}$. The binding examples below make this bookkeeping explicit.

The statistical mechanics protocol

Statistical mechanics assigns probabilities to states by combining their energies with the number of microscopic ways in which they can occur. Here, a relevant state is a macrostate such as "bound" or "unbound," which may contain many distinct microstates. The general protocol is:

1. Identify the relevant states.
2. Define the energy E_i of each state.
3. Count the number of microscopic realizations of each state. This number is its multiplicity, denoted by Ω_i.
4. Compute the statistical weight of each state:
$$w_i = \Omega_i e^{-E_i/(k_B T)}.$$
5. Normalize the weights to obtain the probability of state i:
$$p_i = \frac{w_i}{\sum_j w_j} = \frac{w_i}{Z},$$
where $Z = \sum_j w_j$ is the partition function.

The ligand binding examples below apply this protocol by identifying bound and unbound states, assigning their energies, counting their multiplicities, and normalizing their statistical weights.

5.2 Shannon entropy and the maximum-entropy principle

Shannon entropy provides a measure of the uncertainty associated with a probability distribution.

For a discrete probability distribution

$$\{p_1, p_2, \ldots, p_n\},$$

the Shannon entropy is

$$\boxed{S(\{p_i\}) = -\sum_{i=1}^{n} p_i \ln p_i.}$$

A distribution concentrated entirely in one state has no uncertainty and therefore zero entropy. A broader distribution represents greater uncertainty.

When no constraints other than normalization are imposed, entropy is maximized by assigning equal probability to every available state. This motivates the maximum-entropy principle: choose the probability distribution that satisfies the known constraints while otherwise introducing as little additional structure as possible [20]. In this sense, the maximum-entropy distribution is maximally non-committal about information that has not been specified.

Normalization alone gives equal probabilities on a finite state set. An additional mean constraint changes those probabilities. The full constrained calculation is given in appendix B.5. The dimensionless Shannon entropy used here becomes thermodynamic entropy after multiplication by k_B for the appropriate equilibrium ensemble. A Lagrange multiplier becomes the physical inverse temperature only when the constrained observable is energy.

5.3 Ligand binding by counting

Ligand binding can also be derived by explicitly counting the microscopic states available to bound and unbound molecules. This provides a statistical mechanical route to the receptor-occupancy relation in Eqn. (2), which was previously obtained from mass-action kinetics.

We use the following notation:

- Ω: number of available positions for an unbound ligand;
- L: number of ligand molecules;
- $\epsilon_b < 0$: energy of a ligand in the bound state relative to the chosen reference state;
- k_B: Boltzmann constant;
- T: absolute temperature;
- $\beta = 1/(k_B T)$;
- $W(L)$: number of ways of arranging L indistinguishable ligands among Ω available solution positions.

The lattice is an idealized counting device. Each solution position is assumed to hold at most one ligand, and all solution positions are assigned the same energy unless stated otherwise. For this lattice model,

$$W(L) = \binom{\Omega}{L}.$$

5.3.1 From a volume to a lattice model

Consider a volume containing ligand molecules and receptors.

The continuous three-dimensional volume can be coarse-grained into a lattice of small volume elements. For counting states, the same idea can be represented even more simply as a one-dimensional array of Ω boxes.

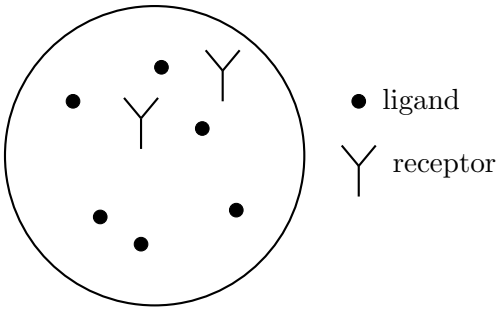


Figure 6: A schematic volume containing ligands and receptors.

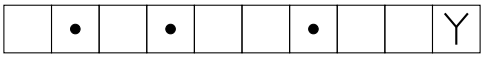


Figure 7: A simplified lattice representation of ligands and a receptor.

i) Two boxes and one ligand

First consider the simplest possible system. There is one ligand and one receptor, and the ligand can either remain unbound or occupy the receptor-binding state.

Choose the energy of the unbound state as the reference,

$$\epsilon_{\text{unbound}} = 0.$$

The bound state has energy

$$\epsilon_{\text{bound}} = \epsilon_b,$$

with $\epsilon_b < 0$ for energetically favorable binding. The Boltzmann weights of the two states are therefore

$$w_{\text{unbound}} = 1,$$

and

$$w_{\text{bound}} = e^{-\beta\epsilon_b}.$$

The partition function is

$$Z = 1 + e^{-\beta\epsilon_b}.$$

The probability of being bound is therefore

$$p_b = \frac{e^{-\beta\epsilon_b}}{1 + e^{-\beta\epsilon_b}}.$$

Equivalently,

$$\frac{p_b}{1 - p_b} = e^{-\beta\epsilon_b}.$$

ii) Many boxes and one ligand

Now suppose that an unbound ligand can occupy any of Ω equivalent solution positions. There are therefore Ω distinct unbound microstates with energy zero, but only one bound state with energy ϵ_b.

The total statistical weight of the unbound state is

$$W_{\text{unbound}} = \Omega.$$

The statistical weight of the bound state is

$$W_{\text{bound}} = e^{-\beta\epsilon_b}.$$

The partition function is

$$Z = \Omega + e^{-\beta\epsilon_b}.$$

Therefore,

$$p_b = \frac{e^{-\beta\epsilon_b}}{\Omega + e^{-\beta\epsilon_b}}.$$

This can be rewritten as

$$p_b = \frac{1}{1 + \Omega e^{\beta\epsilon_b}}.$$

Since

$$\Omega = e^{\ln \Omega},$$

we can write

$$p_b = \frac{1}{1 + \exp\left[\beta\left(\epsilon_b + k_B T \ln \Omega\right)\right]}.$$

The quantity

$$\Delta F = \epsilon_b + k_B T \ln \Omega$$

can be interpreted as the free-energy difference between the bound and unbound macrostates.

The energetic contribution $\epsilon_b < 0$ favors binding. The term $k_B T \ln \Omega > 0$ represents the entropic cost of restricting a ligand that could otherwise occupy many different positions to a single bound state. Binding therefore reflects a competition between energy and entropy.

iii) Many boxes and many ligands

Now consider L indistinguishable ligands and one receptor. If the receptor is unbound, all L ligands occupy solution positions. The number of corresponding microstates is

$$W(L) = \binom{\Omega}{L}.$$

If the receptor is bound, one ligand occupies the receptor and the remaining $L-1$ ligands occupy solution positions. The number of corresponding solution microstates is therefore

$$W(L-1) = \binom{\Omega}{L-1}.$$

The statistical weight of the unbound macrostate is

$$W_{\text{unbound}} = W(L).$$

The statistical weight of the bound macrostate is

$$W_{\text{bound}} = W(L-1)e^{-\beta\epsilon_b}.$$

Therefore,

$$p_b = \frac{W(L-1)e^{-\beta\epsilon_b}}{W(L) + W(L-1)e^{-\beta\epsilon_b}}.$$

The ratio of multiplicities is

$$\frac{W(L-1)}{W(L)} = \frac{L}{\Omega - L + 1}.$$

For a dilute system in which $L \ll \Omega$,

$$\frac{W(L-1)}{W(L)} \approx \frac{L}{\Omega}.$$

Hence

$$p_b \approx \frac{\frac{L}{\Omega}e^{-\beta\epsilon_b}}{1 + \frac{L}{\Omega}e^{-\beta\epsilon_b}}.$$

The ratio L/Ω plays the role of ligand concentration in the lattice model. Thus increasing ligand concentration increases receptor occupancy.

This statistical-mechanical derivation produces the same general saturating form that appeared earlier in the mass-action treatment of ligand–receptor binding.

Homework: active and inactive receptors

Extend the binding-by-counting argument to a receptor that can exist in both an active and an inactive state. The receptor state introduces additional microstates and energetic contributions that must be included in the statistical weights.

5.4 Two binding sites, interactions, and cooperativity

The counting framework can also include interactions between receptors or between multiple binding sites on the same molecular complex. Consider two binding sites and L ligands. Let ϵ_{sol} denote the energy of an unbound ligand in solution, ϵ_b the binding energy of one ligand, and ϵ_{int} an interaction energy that appears when both sites are occupied. Table 3 enumerates the four possible occupancy states and gives their energies, multiplicities, and statistical weights.

State		Energy	Multiplicity	Statistical weight
Y	Y	$L\epsilon_{\text{sol}}$	$W(L)$	$W(L)e^{-\beta L\epsilon_{\text{sol}}}$
Y°	Y	$(L-1)\epsilon_{\text{sol}} + \epsilon_b$	$W(L-1)$	$W(L-1)e^{-\beta[(L-1)\epsilon_{\text{sol}}+\epsilon_b]}$
Y	Y°	$(L-1)\epsilon_{\text{sol}} + \epsilon_b$	$W(L-1)$	$W(L-1)e^{-\beta[(L-1)\epsilon_{\text{sol}}+\epsilon_b]}$
Y°	Y°	$(L-2)\epsilon_{\text{sol}} + 2\epsilon_b + \epsilon_{\text{int}}$	$W(L-2)$	$W(L-2)e^{-\beta[(L-2)\epsilon_{\text{sol}}+2\epsilon_b+\epsilon_{\text{int}}]}$

Table 3: States, energies, multiplicities, and statistical weights for two interacting binding sites.

As shown in Table 3, there are two microscopically distinct singly bound states. If these are grouped into one macrostate, their combined statistical weight is

$$2W(L-1)e^{-\beta[(L-1)\epsilon_{\text{sol}}+\epsilon_b]}.$$

The doubly bound state contains the additional interaction energy ϵ_{int}. If $\epsilon_{\text{int}} < 0$, simultaneous occupancy is energetically favored, producing positive cooperativity. If $\epsilon_{\text{int}} > 0$, simultaneous occupancy is energetically penalized, producing negative cooperativity.

5.5 Case study: reading a Bicoid concentration gradient

Bicoid is a transcription factor that forms a concentration gradient along the anterior–posterior axis of the early *Drosophila* embryo [21]. Bicoid regulates target genes such as *hunchback*, and the local Bicoid concentration helps determine where these genes are expressed [22, 23]. This provides a biological example in which a spatial concentration profile is translated through concentration-dependent regulatory binding into a spatial pattern of gene expression.

5.6 Allostery: the second secret of life

The word *allostery* comes from the Greek *allos*, meaning "other", and *stereos*, meaning "solid" or "structure". In an allosteric protein, a regulatory molecule binds at a site that is physically distinct from the protein's active or functional site. Binding at the regulatory site changes the energetic or conformational state of the protein and thereby alters activity at the other site [24]. The central idea is therefore coupling between spatially distinct sites on the same molecular system.

Allosteric regulation of this kind is widespread in biology. Metabolic pathways provide many examples in which metabolites regulate enzymes upstream or downstream in the pathway [24]. Glycolysis, which converts glucose into pyruvate, contains several such regulatory steps.

Allostery provides a molecular mechanism through which a protein can sense one chemical signal and alter its response to another.

5.6.1 Statistical mechanics of allosteric enzymes

A useful framework for describing allosteric enzymes is the Monod-Wyman-Changeux (MWC) model [25]. In this picture, an enzyme can fluctuate between different conformational states, commonly an active state and an inactive state. These states can have different substrate-binding affinities and different catalytic activities. The relative probabilities of the states, together with ligand binding, determine the overall activity of the enzyme.

The same states-and-weights idea used for ligand binding can therefore be extended to allostery. Instead of considering only bound and unbound states, we enumerate combinations of conformational state and ligand occupancy, assign each state an appropriate statistical weight, and normalize over all possible states.

Einav, Mazutis, and Phillips developed this states-and-weights treatment into a statistical-mechanical framework for allosteric enzymes [26]. Their formulation first recasts Michaelis–Menten kinetics in terms of enzyme states and statistical weights and then extends the framework to MWC enzymes, allosteric regulators, competitive inhibitors, and enzymes with multiple substrate-binding sites.

6 Finding useful representations

Choosing a variable was already part of constructing a model in section 2. The question now is how to represent a function, dataset, or biological shape so that its important structure becomes visible. Fourier modes offered one example; normal modes and principal components offer others.

The way in which a problem is represented can determine which aspects of the problem become easy or difficult to see. A useful representation exposes the structure that matters while suppressing details that are irrelevant to the question being asked.

6.1 Vectors as representations of biological objects

Many complicated objects can be represented as vectors. Once an object has been written as a vector, the tools of linear algebra can be used to compare objects, transform representations, and identify important directions of variation.

Examples include:

- an image, represented by a vector of pixel intensities;
- a transcriptome, represented by a vector of gene-expression levels;
- a biological shape, represented by coordinates or by coefficients in a suitable basis, such as spherical harmonics or finite elements;
- a probability distribution, represented by the probabilities associated with each possible state.

The Buri probability vector is developed alongside its experiment in section 3.4.

6.2 Approximating functions with simpler building blocks

A complicated function or collection of data can often be represented approximately using simpler mathematical objects. Different representations are useful in different situations.

Taylor expansions use local derivatives; interpolation uses values at known points. Their full explanations, worked examples, and interpolation figure are given in appendices A.3 and A.4.

6.2.1 ReLU and piecewise-linear representations

A rectified linear unit, or ReLU, is a piecewise-linear function defined by

$$\boxed{\mathrm{ReLU}(x) = \max(0, x).}$$

Equivalently,

$$\mathrm{ReLU}(x) = \begin{cases} 0, & x < 0, \\ x, & x \geq 0. \end{cases}$$

The function is therefore flat for negative inputs and linear for positive inputs.

ReLU functions are widely used as activation functions in neural networks. Although a single ReLU is extremely simple, combinations of many ReLU units can produce complicated piecewise-linear functions. Thus ReLUs provide another way of building a complicated function from simple components.

6.3 Least squares: choosing a representation from data

Least squares is a method for finding a model that best represents a collection of data.

Suppose we have measurements

$$(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)$$

and a model that predicts values

$$\hat{y}_i = f(x_i).$$

The residual for measurement i is the difference between the observed and predicted values,

$$r_i = y_i - \hat{y}_i.$$

Least squares chooses the parameters of the model so that the sum of the squared residuals is as small as possible,

$$\boxed{\text{minimize} \qquad \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 .}$$

Squaring the residuals prevents positive and negative errors from simply cancelling one another. The result is the model that gives the smallest total squared discrepancy from the data according to this criterion.

Legendre, Gauss, and least squares

Adrien-Marie Legendre published the method of least squares in 1805 as a practical method for extracting parameters from astronomical observations when there were more measurements than unknown quantities. Carl Friedrich Gauss published his treatment in 1809 and claimed that he had used the method earlier, producing a lasting historical priority dispute between Gauss and Legendre [27]. The method subsequently became fundamental for fitting mathematical models to noisy observations.

These methods provide different ways of replacing something complicated by something simpler. A Taylor expansion represents a function locally using a polynomial. Linear interpolation represents the function between two known points using a straight line. ReLU functions provide simple piecewise-linear building blocks from which more complicated functions can be constructed. Least squares chooses the representation that best matches a collection of measurements according to a specified error measure.

6.3.1 A straight-line fitting problem

Suppose we have N data points

$$(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)$$

and wish to represent them by a straight line,

$$y = ax + b.$$

For each data point, the residual between the observed and predicted value is

$$r_i = y_i - (ax_i + b).$$

The least-squares method chooses a and b to minimize the sum of the squared residuals,

$$E(a, b) = \sum_{i=1}^{N} [y_i - (ax_i + b)]^2 .$$

This provides a quantitative definition of the "best-fitting" line.

6.3.2 Why squares? Gaussian measurement noise

One reason squared errors arise naturally is that measurement errors are often modeled as independent Gaussian random variables.

Suppose

$$y_i = ax_i + b + \epsilon_i,$$

where

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2).$$

The probability density of observing y_i for given a and b is then

$$p(y_i \mid x_i, a, b) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - ax_i - b)^2}{2\sigma^2}\right].$$

Assuming that the errors of different measurements are independent, the likelihood of the complete dataset is

$$\mathcal{L}(a, b) = \prod_{i=1}^{N} p(y_i \mid x_i, a, b).$$

Taking the logarithm converts the product into a sum,

$$\ln \mathcal{L}(a, b) = \text{constant} - \frac{1}{2\sigma^2} \sum_{i=1}^{N} [y_i - (ax_i + b)]^2 .$$

Maximizing the likelihood is therefore equivalent to minimizing

$$\sum_{i=1}^{N} [y_i - (ax_i + b)]^2 .$$

Thus least squares follows naturally from a model in which the measurement noise is independent and Gaussian with constant variance.

The parameter calculation, normal equations, matrix representation, and explicit slope and intercept are retained in appendix B.4.

6.4 Finding the right coordinates

A representation can be formally correct and still be inconvenient. An important strategy in physics is therefore to search for coordinates in which the important structure of the problem becomes simple.

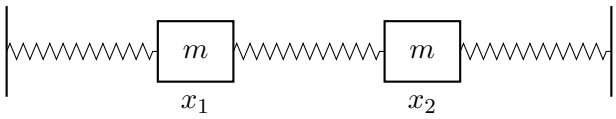


Figure 8: Two coupled oscillators. The individual displacements x_1 and x_2 provide one coordinate system, while the normal modes provide another.

6.4.1 Coupled oscillators and normal modes

Consider two masses coupled by springs.

The individual mass displacements x_1 and x_2 are natural coordinates, but their equations of motion are coupled.

For a symmetric system, more useful coordinates are the combinations

$$q_+ = \frac{x_1 + x_2}{\sqrt{2}}$$

and

$$q_- = \frac{x_1 - x_2}{\sqrt{2}}.$$

The coordinate q_+ describes the mode in which the masses move together. The coordinate q_- describes the mode in which they move oppositely. These are the normal modes of the system. In normal-mode coordinates, the coupled problem can be decomposed into independent motions. This illustrates the importance of finding coordinates adapted to the structure of the problem.

6.4.2 Covariance: which variables fluctuate together?

The variance of a variable x measures its fluctuations around its mean,

$$\mathrm{Var}(x) = \left\langle (x - \langle x \rangle)^2 \right\rangle.$$

Expanding the square gives

$$\mathrm{Var}(x) = \langle x^2 \rangle - \langle x \rangle^2.$$

For two variables x_1 and x_2, the covariance is

$$\mathrm{Cov}(x_1, x_2) = \langle (x_1 - \langle x_1 \rangle)(x_2 - \langle x_2 \rangle) \rangle.$$

The covariance matrix for two variables is

$$C = \begin{pmatrix} \left\langle (x_1 - \langle x_1 \rangle)^2 \right\rangle & \langle (x_1 - \langle x_1 \rangle)(x_2 - \langle x_2 \rangle) \rangle \\ \langle (x_1 - \langle x_1 \rangle)(x_2 - \langle x_2 \rangle) \rangle & \left\langle (x_2 - \langle x_2 \rangle)^2 \right\rangle \end{pmatrix}.$$

The diagonal entries describe the variance of each coordinate. The off-diagonal entries describe how fluctuations in different coordinates are correlated.

6.4.3 Principal component analysis

Principal component analysis, or PCA, provides a systematic way to find coordinates that capture the dominant variation in a dataset [28, 29].

For variables collected into a vector $\mathbf{x}$, define the covariance matrix

$$C_{ij} = \langle (x_i - \langle x_i \rangle)\,(x_j - \langle x_j \rangle) \rangle\,.$$

The principal components are the eigenvectors of this matrix,

$$C\mathbf{v}_\mu = \lambda_\mu \mathbf{v}_\mu.$$

The eigenvectors $\mathbf{v}_\mu$ define new coordinate directions. The corresponding eigenvalues λ_μ measure the variance of the data along those directions.

The first principal component is the direction along which the data vary most. The second is the orthogonal direction containing the next-largest variance, and so on.

If only a few eigenvalues are large, a high-dimensional dataset can be represented accurately using only a few principal components. PCA therefore provides both a change of coordinates and a method for dimensionality reduction.

6.5 The eigenworm: low-dimensional shape and behavior

A biological example of finding useful coordinates is the quantitative analysis of *Caenorhabditis elegans* posture and locomotion by Stephens et al. [30]. Their goal was to determine whether the apparently complicated set of worm postures could be represented using only a few collective coordinates.

A simplified image-analysis and PCA pipeline is:

0. Parse the image data.
1. Segment the worm from the image background.
2. Skeletonize the worm to determine its centerline.
3. Fit or sample a smooth contour along the centerline.
4. Compute the tangent angle along the body for every recorded time.
5. Arrange the tangent-angle data as a function of body position and time.
6. Compute the covariance matrix of the angular fluctuations.
7. Perform principal component analysis.
8. Decompose each observed posture into the resulting eigenmodes.

Let $\theta(s,t)$ denote the tangent angle at position s along the worm body at time t. After removing the overall orientation of the worm, a posture can be represented by the set of angles along its body (Fig 9).

The covariance between two body positions s and s' is

$$C(s,s') = \left\langle \left[\theta(s,t) - \langle \theta(s,t) \rangle_t\right] \left[\theta(s',t) - \langle \theta(s',t) \rangle_t\right] \right\rangle_t,$$

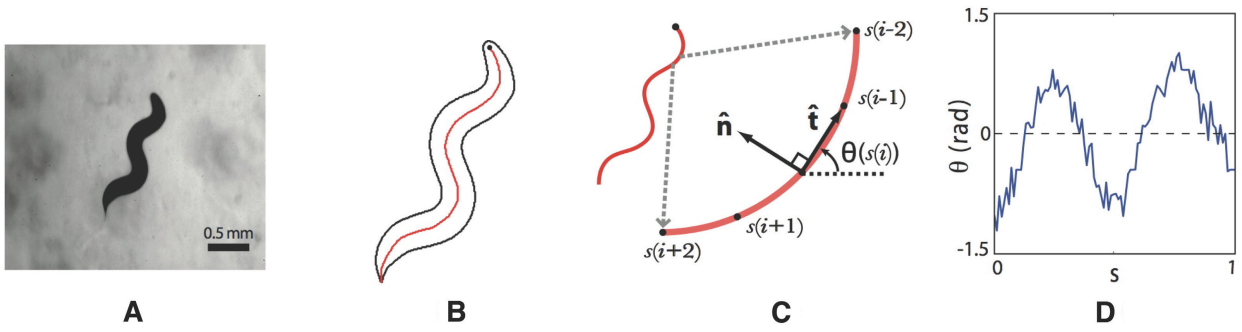


Figure 9: Representing the shape of *C. elegans*. A raw microscopy image of the worm is reduced to a curve along its centerline. Position along the body is described by the normalized arclength s, while the local tangent angle $\theta(s)$ records the orientation of each body segment. Subtracting the mean angle removes the overall orientation of the animal, producing an intrinsic description of posture that can be compared across images. Adapted from Figure 1 of Stephens et al. [30]; panels have been rearranged.

where $\langle\cdot\rangle_t$ denotes an average over the recorded postures. The eigenvectors of this covariance matrix satisfy

$$\sum_{s'} C(s,s')u_\mu(s') = \lambda_\mu u_\mu(s).$$

These principal components are called *eigenworms*. The worm shape can then be represented approximately as

$$\theta(s,t) \approx \sum_{\mu=1}^{K} a_\mu(t)u_\mu(s).$$

Stephens et al. found that four eigenworms account for approximately 95% of the observed variance in worm shapes (Fig. 10) [30].

Thus an object initially represented using roughly one hundred local angular coordinates can be represented, to good approximation, by only four mode amplitudes,

$$\mathbf{a}(t) = \begin{pmatrix} a_1(t) \\ a_2(t) \\ a_3(t) \\ a_4(t) \end{pmatrix}.$$

The important result is not merely data compression. The new coordinates provide a compact description of the behavior and dynamics of the animal.

6.5.1 Geometric phase and locomotion

Rieser et al. used a geometric-phase framework to connect cyclic changes in body shape to locomotion in undulating animals operating in highly damped environments [31]. Periodic locomotion can be represented as a closed trajectory through a low-dimensional space of body shapes. The coordinates of this space describe how much of each shape mode is present at a given time.

As an animal cycles through a sequence of shapes, it traces a loop through shape space. In the friction-dominated regimes considered in the study, the geometry of this loop predicts the

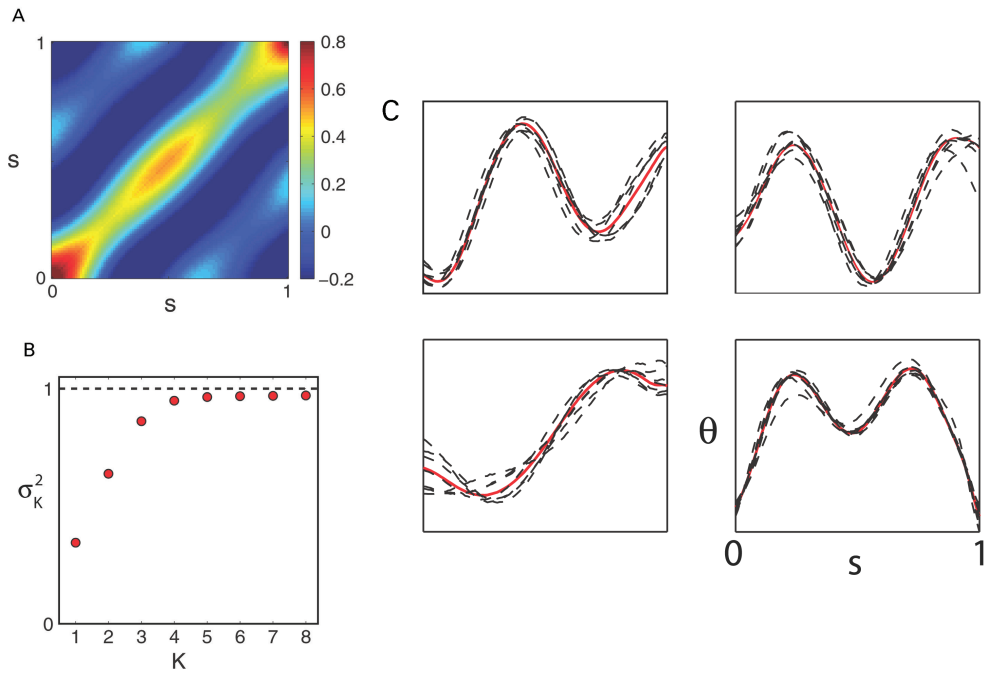


Figure 10: Covariance, dimensionality, and the eigenworms. (A) The covariance matrix $C(s, s')$ measures how tangent-angle fluctuations at different positions along the worm body vary together. Its smooth structure indicates that the observed posture fluctuations may be described by a small number of collective modes. (B) The cumulative fraction of posture variance captured by the first K modes rises rapidly; four modes account for approximately 95% of the total variance. (C) The corresponding eigenvectors of the covariance matrix are the eigenworms $u_\mu(s)$. Population-averaged eigenworms are shown in red, while eigenworms calculated for individual animals are shown in black, demonstrating that the dominant modes are reproducible across worms. Adapted from Figure 2A–C of Stephens et al. [30].

net displacement accumulated during a locomotor cycle. Thus an appropriate low-dimensional representation can expose a direct relationship between body-shape dynamics and locomotor performance.

7 The philosophy of physical approaches to biological modeling

The case studies reveal recurring structures: gain and loss, probability flow, state counting, and useful changes of coordinates. The closing questions concern what these descriptions explain and where their assumptions stop being adequate.

7.1 The sameness of things

The same physical ideas appear repeatedly across very different biological systems. Random walks, for example, arise in processes ranging from molecular hopping to the conformations of polymers. Likewise, similar ideas about binding, switching, cooperativity, and feedback appear in many important regulatory systems.

Examples discussed in the course include:

- **Chemotaxis.** Measurements of bacterial chemotaxis reveal highly sensitive receptor responses, while models of interacting two-state receptors show how the same receptor clusters can operate in regimes dominated by sensitivity or cooperativity [32, 33].
- **Quorum sensing.** Measurements of the LuxN receptor in *Vibrio harveyi* show how a quantitative receptor model can be used to infer signaling parameters from in vivo responses [34].
- **Gene expression.** Single-cell measurements of gene expression show how stochastic production and degradation of molecules generate fluctuations and transcriptional bursts [15].

Would life look different elsewhere?

The distinction between general organizing principles and contingent molecular implementations was introduced in section 1. After the case studies, the question can be revisited more concretely: which explanations depend on particular molecules, and which depend on structures such as autocatalysis, stochastic sampling, or energetic coupling?

7.2 Defiance: maintaining living systems away from equilibrium

One way to think about living systems is through their ability to maintain states that would not persist at thermodynamic equilibrium. They do so by coupling internal processes to continual inputs of free energy, thereby breaking detailed balance (every microscopic transition is balanced by its reverse, so there are no net probability currents) and sustaining nonequilibrium dynamics [35].

Examples include:

- creating and maintaining concentration gradients;
- increasing biochemical specificity through energy-driven kinetic proofreading [36];
- adapting to changes in the environment;
- enforcing ordered kinetic sequences, in which a later step cannot occur before a required earlier step ("socks before shoes" kinetics);
- sustaining organized structures that persist only while energy is continually supplied and dissipated.

Living systems therefore do not violate thermodynamics. They maintain nonequilibrium organization through continual free-energy input and dissipation [35].

7.3 What physical approaches bring to biology

Physical approaches to biology are not defined only by the use of equations or quantitative measurements. They also provide particular ways of deciding what constitutes an explanation or an understanding of a biological system, such as:

- providing alternative definitions of what it means to understand a phenomenon;
- exposing hidden sameness between apparently unrelated biological systems;
- providing a different perspective on what is meant by a mechanism;
- allowing null hypotheses to be formulated explicitly and quantitatively;
- forcing us to state clearly what is known and what remains unknown, in the spirit of what Joshua Epstein calls "militant ignorance" [37];
- emphasizing conceptual knowledge rather than the accumulation of isolated facts.

A recurring theme throughout the course is that the choice of representation, model, or null hypothesis is itself part of the scientific reasoning. The aim is not merely to calculate a quantity, but to identify a description in which the essential structure of the biological problem becomes visible.

7.3.1 Dimensionless numbers as quantitative perspective

A final example of the physical approach is the construction of dimensionless numbers. By comparing two effects or scales so that their units cancel, these ratios replace absolute magnitude with relative importance. Values much smaller than one, of order one, or much larger than one identify regimes in which the denominator dominates, the two effects are comparable, or the numerator dominates, respectively.

A well-chosen dimensionless number can reveal which effects govern a system, which can be neglected, and which apparently different phenomena share the same underlying structure. Its significance therefore lies less in its name or precise value than in the choice of quantities being compared. In this way, dimensionless numbers exemplify a central contribution of physical thinking to biology: finding representations that make otherwise abstract relationships quantitatively intelligible.

The following dimensionless number illustrate this spirit. Some of these numbers aren't canonically named, but they are useful to think about.

Biologically relevant examples

Dimensionless number	**Ratio**
Reynolds number	$\mathrm{Re} = \dfrac{\text{inertial force}}{\text{viscous force}} = \dfrac{u\rho L}{\mu}$
Péclet number	$\mathrm{Pe} = \dfrac{\text{diffusion time}}{\text{advection time}} = \dfrac{L^2/D}{L/V}$
Jesus number	$\mathrm{Je} = \dfrac{\text{surface-tension force}}{\text{gravitational force}} = \dfrac{\gamma\ell}{\rho\ell^3 g}$
Langmuir number	$\mathrm{La} = \dfrac{\text{binding energy}}{\text{entropy loss of ligands}} = \dfrac{\epsilon_b}{k_B T \ln\frac{c}{c_0}}$
Bjerrum number	$\mathrm{Bj} = \dfrac{\text{electrostatic energy}}{\text{thermal energy}} = \dfrac{e^2}{4\pi\epsilon_0 D\ell k_B T}$
Hooke number	$\mathrm{Ho} = \dfrac{\text{elastic energy stored in a spring}}{\text{thermal energy}} = \dfrac{k\ell^2}{2k_B T}$

Anthropogenic examples

Dimensionless number	**Ratio**
Humanification number	$\mathrm{Hu} = \dfrac{\text{anthropomass}}{\text{biomass}}$
Keeling number	$\mathrm{Ke} = \dfrac{\mathrm{CO_2}\text{ added by humans}}{\text{total }\mathrm{CO_2}\text{ before humans}}$
Wildlife number	$\mathrm{Wi} = \dfrac{\text{biomass of humans + livestock}}{\text{biomass of wild animals}}$
Haber number	$\mathrm{Ha} = \dfrac{\mathrm{N_2}\text{ flux through Haber-Bosch process}}{\mathrm{N_2}\text{ flux through biological processes}}$
Amazon number	$\mathrm{Am} = \dfrac{\text{daily human water use}}{\text{Amazon drainage rate}}$
Mortality number	$\mathrm{Mo} = \dfrac{\text{accidental deaths in the US}}{\text{annual deaths in the US}}$

Across these examples, the physical approach does not provide a single theory of life, but a disciplined way of asking questions about living systems. It begins by estimating scales, choosing variables and representations, stating assumptions, and constructing the simplest model capable of making a testable prediction. Means, distributions, random walks, state counting, and changes of coordinates are useful not because living systems are simple, but because they make precise what would otherwise remain qualitative or ambiguous. A model is therefore neither a miniature copy of biology nor an end in itself; it is a tool for exposing what is known, what has been assumed, and what remains unexplained. This combination of quantitative clarity, conceptual economy, and explicit ignorance is the enduring contribution of physical thinking to biology.

A Mathematical methods

A.1 Reading mathematical notation

A.1.1 Indices, sums, and products

An index labels members of a collection. In x_i, the index i might label a cell, measurement, spatial site, or component of a vector. The summation notation

$$\sum_{i=1}^{N} x_i = x_1 + x_2 + \cdots + x_N$$

means to add the terms obtained by letting i run from 1 to N. Similarly,

$$\prod_{i=1}^{N} x_i = x_1 x_2 \cdots x_N.$$

Indices are dummy labels inside a complete sum. For example, $\sum_{i=1}^{N} x_i = \sum_{j=1}^{N} x_j$. The symbol used for the index changes, but the value does not.

Worked example: a sample mean

For measurements $x_1 = 2$, $x_2 = 5$, and $x_3 = 8$, the sample mean is

$$\bar{x} = \frac{1}{3}\sum_{i=1}^{3} x_i = \frac{2+5+8}{3} = 5.$$

A.1.2 Powers, exponentials, and logarithms

The exponential function e^x is special because its derivative equals itself. It converts addition in the exponent into multiplication:

$$e^{a+b} = e^a e^b.$$

The natural logarithm $\ln x$ is its inverse:

$$\ln(e^x) = x, \qquad e^{\ln x} = x \quad (x > 0).$$

Useful logarithm rules are

$$\ln(xy) = \ln x + \ln y, \qquad \ln\left(\frac{x}{y}\right) = \ln x - \ln y, \qquad \ln(x^a) = a \ln x.$$

Exponential growth has the form $M(t) = M_0 e^{kt}$ with $k > 0$. Its doubling time t_2 satisfies $M(t_2) = 2M_0$, so

$$t_2 = \frac{\ln 2}{k}.$$

Exponential decay has the form $A(t) = A_0 e^{-t/\tau}$. The time τ is the time at which the amplitude has fallen to $1/e$ of its initial value.

A.2 Calculus: change, accumulation, and several variables

Calculus provides two complementary operations. A derivative measures a local rate of change. An integral measures accumulated change or total amount. The fundamental theorem of calculus connects them.

A.2.1 Derivatives as local rates of change

For a function $y = f(t)$, the derivative is

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \lim_{\Delta t \to 0} \frac{f(t + \Delta t) - f(t)}{\Delta t}.$$

The fraction before the limit is an average rate of change over a finite interval. The derivative is the rate in the limit of an arbitrarily short interval.

Some basic derivative rules are

$$\frac{\mathrm{d}}{\mathrm{d}x} x^n = n x^{n-1}, \qquad \frac{\mathrm{d}}{\mathrm{d}x} e^{ax} = a e^{ax}, \qquad \frac{\mathrm{d}}{\mathrm{d}x} \ln x = \frac{1}{x}.$$

The product and chain rules are

$$\frac{\mathrm{d}}{\mathrm{d}x}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x),$$

$$\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = f'(g(x))g'(x).$$

Worked example: verify exponential relaxation

Suppose a quantity relaxes toward a steady value m_{ss} over a timescale τ:

$$m(t) = m_{\mathrm{ss}} + (m_0 - m_{\mathrm{ss}})e^{-t/\tau}.$$

Differentiating gives

$$\frac{\mathrm{d}m}{\mathrm{d}t} = -\frac{m_0 - m_{\mathrm{ss}}}{\tau} e^{-t/\tau}.$$

The displacement from steady state is $m(t) - m_{\mathrm{ss}} = (m_0 - m_{\mathrm{ss}})e^{-t/\tau}$. Therefore

$$\frac{\mathrm{d}m}{\mathrm{d}t} = -\frac{m - m_{\mathrm{ss}}}{\tau},$$

which says directly that the system moves toward steady state at a rate proportional to its remaining displacement.

A.2.2 Partial derivatives

A function of several variables can change in several independent directions. For $P = P(x, t)$, $\partial P/\partial t$ measures change with time while holding position fixed, whereas $\partial P/\partial x$ measures spatial change at fixed time. The second spatial derivative $\partial^2 P/\partial x^2$ measures how the slope changes with position. At a smooth local maximum it is typically negative, and at a smooth local minimum it is typically positive. This explains qualitatively why the diffusion equation lowers peaks and fills valleys.

A.2.3 Integrals as accumulated quantities

The definite integral $\int_a^b f(x)\,\mathrm{d}x$ is the signed area accumulated under f between a and b. If $F'(x) = f(x)$, the fundamental theorem of calculus gives

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Worked example: normalize an exponential density

For $p(t) = \lambda e^{-\lambda t}$ with $t \geq 0$ and $\lambda > 0$,

$$\int_0^\infty p(t)\,\mathrm{d}t = \Big[-e^{-\lambda t}\Big]_0^\infty = 1.$$

A.2.4 The continuity equation

Let $\rho(x,t)$ be an amount per unit length and $J(x,t)$ the amount crossing a point per unit time. Conservation implies

$$\frac{\partial \rho}{\partial t} = -\frac{\partial J}{\partial x}.$$

Integrating over an interval gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_a^b \rho(x,t)\,\mathrm{d}x = J(a,t) - J(b,t).$$

A.3 Taylor expansions and controlled approximations

A Taylor expansion replaces a function near a chosen point by a polynomial whose value and derivatives match those of the function. For expansion around x_0,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots.$$

The notation $\mathcal{O}((x - x_0)^2)$ denotes an omitted error of second order or smaller as $x \to x_0$.

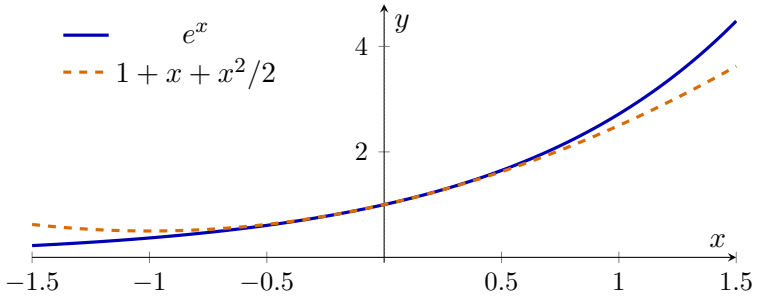


Figure 11: The quadratic Taylor approximation to e^x agrees with the function near the expansion point and gradually departs from it farther away

Common expansions are

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots, \qquad \sin x = x - \frac{x^3}{3!} + \cdots,$$

$$\cos x = 1 - \frac{x^2}{2!} + \cdots, \qquad \ln(1 + x) = x - \frac{x^2}{2} + \cdots.$$

Worked example: the continuum limit

Expanding $P(x+a,t)$ and $P(x-a,t)$ around x and adding gives

$$P(x+a,t)+P(x-a,t)-2P(x,t)=a^2\frac{\partial^2 P}{\partial x^2}+\mathcal{O}(a^4).$$

Odd powers cancel because the walk is symmetric.

Common mistakes

- State the small quantity, such as $a/L \ll 1$.
- A Taylor approximation is local, not automatically global.
- Smoothness supports finite expansions with remainders, but an infinite Taylor series need not equal every smooth function.

A.4 Linear interpolation

For known points (x_1,y_1) and (x_2,y_2), linear interpolation gives

$$y(x)=y_1+\frac{y_2-y_1}{x_2-x_1}(x-x_1).$$

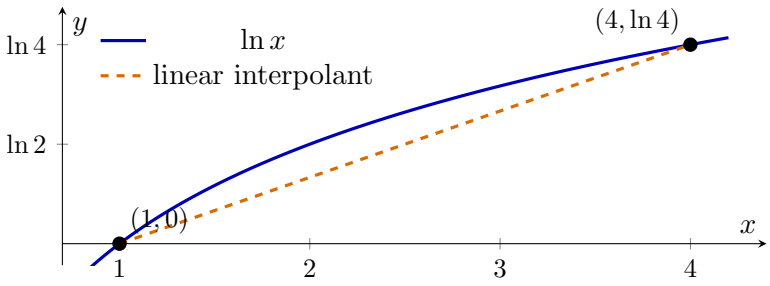


Figure 12: Linear interpolation of the concave function $\ln x$ between two known points. The interpolating line lies below the function between the endpoints

Worked example

For $y=\ln x$, use the known points $(1,0)$ and $(4,\ln 4)$. The interpolating line is

$$y_{\text{lin}}(x)=\frac{\ln 4}{3}(x-1).$$

At $x=2$, it gives $y_{\text{lin}}(2)=\ln 4/3\approx 0.462$, whereas $\ln 2\approx 0.693$. Because $\ln x$ bends downward, its secant line lies below the curve, so this interpolation underestimates the function between the endpoints.

A.5 Ordinary differential equations and relaxation

An ordinary differential equation, or ODE, relates a quantity to derivatives with respect to one independent variable, usually time. Solving an ODE means finding a function that satisfies both the equation and an initial condition.

For $\mathrm{d}y/\mathrm{d}t = ky$, separation of variables and integration give

$$y(t) = y(0)e^{kt}.$$

Positive k gives growth and negative k gives decay.

For production and first-order loss,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = A - By,$$

the steady state and full solution are

$$y_{\mathrm{ss}} = \frac{A}{B}, \qquad y(t) = \frac{A}{B} + \left[y(0) - \frac{A}{B}\right] e^{-Bt}.$$

The relaxation time is $\tau = 1/B$.

Worked example: mRNA turnover

For $r = 6\,\mathrm{mRNA\,h^{-1}}$, $\gamma = 2\,\mathrm{h^{-1}}$, and $m(0) = 0$,

$$m(t) = 3\left(1 - e^{-2t}\right).$$

After one relaxation time, the system has completed about 63% of the change to steady state.

Linearizing near a steady state

For an autonomous equation $\mathrm{d}y/\mathrm{d}t = f(y)$, a steady state y_* satisfies $f(y_*) = 0$. To determine what happens near the steady state, write the solution as $y(t) = y_* + \delta y(t)$, where δy is a small displacement from y_*. A Taylor expansion gives

$$f(y_* + \delta y) = f(y_*) + f'(y_*)\delta y + \mathcal{O}(\delta y^2).$$

Because $f(y_*) = 0$, neglecting terms quadratic and higher order in the small displacement gives

$$\frac{\mathrm{d}\,\delta y}{\mathrm{d}t} \approx f'(y_*)\delta y.$$

This replacement of the original nonlinear equation by a linear approximation near the steady state is called *linearization*. Its solution is

$$\delta y(t) = \delta y(0)e^{f'(y_*)t}.$$

If $f'(y_*) < 0$, small displacements decay and the steady state is locally stable. If $f'(y_*) > 0$, small displacements grow and the steady state is locally unstable. If $f'(y_*) = 0$, this linear approximation does not determine the stability, and higher-order terms must be considered.

A.6 Gain-loss bookkeeping and master equations

A master equation is an evolution equation for probabilities. It is built by listing every way probability enters a state and every way it leaves.

Worked example: a lattice random walk

Let $p_n(t)$ be the probability of finding a walker at lattice site n at time t. The walker jumps to either neighboring site at rate k. Site n gains probability from sites $n-1$ and $n+1$ and loses probability through its two outgoing jumps, so

$$\frac{\mathrm{d}p_n}{\mathrm{d}t} = kp_{n-1} + kp_{n+1} - 2kp_n.$$

The first two terms are gains and the last term is the loss. When the equations are summed over all sites, every jump appears once as a loss from one site and once as a gain to another, so total probability remains one.

A.7 Partial differential equations, separation of variables, and boundary conditions

A PDE involves derivatives with respect to more than one independent variable. The diffusion equation

$$\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}$$

requires an initial spatial profile and boundary conditions.

Boundary conditions

The diffusion equation describes how the profile changes inside a spatial domain, but it does not specify what happens at the edges. Boundary conditions supply this missing information. On a one-dimensional interval $0 \le x \le L$, three common choices are the following.

- **Periodic boundaries.** The two ends of the interval are treated as the same point:

  $$P(0,t) = P(L,t), \qquad \left.\frac{\partial P}{\partial x}\right|_{x=0} = \left.\frac{\partial P}{\partial x}\right|_{x=L}.$$

  Probability leaving through one end re-enters through the other, as if the interval had been joined into a ring. Periodic boundaries permit Fourier modes with wave numbers $q_n = 2\pi n/L$.

- **Fixed-value boundaries.** The value of the field is prescribed at each edge:

  $$P(0,t) = P_{\text{left}}, \qquad P(L,t) = P_{\text{right}}.$$

  These are also called *Dirichlet boundary conditions*. For temperature diffusion, they describe ends held at fixed temperatures. For a probability density, the choice $P = 0$ at an edge represents an absorbing boundary: probability reaching that edge leaves the system.

- **Zero-flux boundaries.** Nothing is allowed to cross either edge:

  $$J(0,t) = J(L,t) = 0.$$

  Using Fick's law, $J = -D\,\partial P/\partial x$, this is equivalent to

  $$\left.\frac{\partial P}{\partial x}\right|_{x=0} = \left.\frac{\partial P}{\partial x}\right|_{x=L} = 0.$$

  These are also called *Neumann* or *reflecting* boundary conditions. They conserve the total probability inside the interval.

Boundary conditions determine which spatial modes are allowed, while the initial condition $P(x,0)$ determines the amplitude of each allowed mode.

Trying $P(x,t) = X(x)T(t)$ gives

$$\frac{1}{DT}\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{X}\frac{\mathrm{d}^2X}{\mathrm{d}x^2} = -q^2.$$

Hence

$$T(t) = T_0 e^{-Dq^2 t}, \qquad X(x) = A\cos(qx) + B\sin(qx).$$

Periodic boundaries select $q_n = 2\pi n/L$. Boundary conditions select allowed modes; initial conditions select their amplitudes.

A.8 Vectors, matrices, and systems of linear equations

A vector is an ordered list of components in a chosen coordinate system,

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

The dot product and Euclidean length are

$$\mathbf{x}\cdot\mathbf{y} = \sum_i x_i y_i, \qquad \|\mathbf{x}\| = \sqrt{\mathbf{x}\cdot\mathbf{x}}.$$

Vectors are orthogonal when their dot product is zero.

A matrix maps one vector to another. For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix},$$

we have

$$A\mathbf{x} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

Matrix multiplication is generally not commutative: $AB \neq BA$.

For a linear system $A\mathbf{x} = \mathbf{b}$, the formal solution $\mathbf{x} = A^{-1}\mathbf{b}$ exists when A is invertible, but numerical work generally solves the system directly.

A.9 Eigenvalues, eigenvectors, and eigenfunctions

An eigenvector is a direction preserved by a matrix:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The eigenvalue λ multiplies its amplitude. Eigenvalues satisfy $\det(A - \lambda I) = 0$, where det denotes the determinant and I denotes the identity matrix. For a two-dimensional system,

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The determinant of a 2×2 matrix is

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Therefore,

$$\det(A - \lambda I) = \det \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} = (a - \lambda)(d - \lambda) - bc.$$

Setting this expression equal to zero gives a polynomial equation for λ whose solutions are the eigenvalues of A.

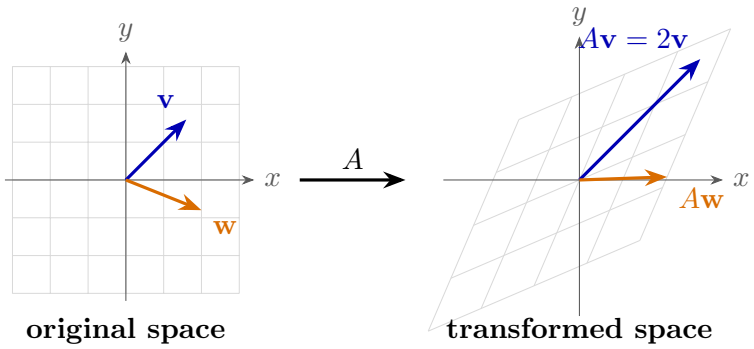


Figure 13: The matrix $A = \left(\begin{smallmatrix} 1.4 & 0.6 \\ 0.6 & 1.4 \end{smallmatrix}\right)$ deforms the coordinate grid. The eigenvector $\mathbf{v} = (0.8, 0.8)^\mathsf{T}$ is stretched without changing direction, while the generic vector $\mathbf{w} = (1, -0.4)^\mathsf{T}$ changes both length and direction

Worked example

For

$$A = \begin{pmatrix} 1.4 & 0.6 \\ 0.6 & 1.4 \end{pmatrix},$$

take the vectors used in Figure 13:

$$\mathbf{v} = \begin{pmatrix} 0.8 \\ 0.8 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 1 \\ -0.4 \end{pmatrix}.$$

Multiplication gives

$$A\mathbf{v} = \begin{pmatrix} 1.6 \\ 1.6 \end{pmatrix} = 2\mathbf{v}, \qquad A\mathbf{w} = \begin{pmatrix} 1.16 \\ 0.04 \end{pmatrix}.$$

Thus $\mathbf{v}$ is an eigenvector: it is stretched by a factor of two without changing direction. The vector $\mathbf{w}$ changes direction and is not an eigenvector.

An eigenfunction f of an operator $\mathcal{L}$ satisfies $\mathcal{L}f = \lambda f$. For example,

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} e^{iqx} = -q^2 e^{iqx}.$$

For a real symmetric matrix such as this one, eigenvectors associated with different eigenvalues are perpendicular.

A.10 Fourier series, superposition, and delta distributions

A Fourier series represents a periodic function as a sum of sines and cosines with different wavelengths. The coefficients state how much of each wave is present in the function. For a periodic domain of length L,

$$f(x) = A_0 + \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{2\pi n x}{L}\right) + B_n \sin\left(\frac{2\pi n x}{L}\right) \right].$$

The coefficients are coordinates measuring each spatial frequency.

Fourier modes are eigenfunctions of translation and differentiation. The wave number and wavelength are

$$q_n = \frac{2\pi n}{L}, \qquad \lambda_n = \frac{L}{n}.$$

Orthogonality permits coefficients to be isolated, for example

$$A_n = \frac{2}{L} \int_0^L f(x) \cos(q_n x)\, \mathrm{d}x.$$

For diffusion, each mode evolves independently as

$$P_n(x,t) = P_n(x,0) e^{-D q_n^2 t}.$$

An $n = 3$ mode decays nine times faster than an $n = 1$ mode on the same domain.

The Dirac delta is defined by

$$\int \delta(x - x_0) f(x)\, \mathrm{d}x = f(x_0).$$

It is an idealized unit amount concentrated at one point, not an ordinary function. A finite Fourier approximation to it can oscillate and need not remain nonnegative everywhere.

A.11 Probability, expectation, and variance

For discrete states, $p_i \geq 0$ and $\sum_i p_i = 1$. The expectation is

$$\langle f \rangle = \sum_i p_i f_i.$$

If $m = 0, 1, 2$ with probabilities 0.2, 0.5, and 0.3, then $\langle m \rangle = 1.1$. No cell needs to contain 1.1 molecules; it is a population mean.

For a continuous variable, a density satisfies

$$p(x) \geq 0, \qquad \int p(x)\, \mathrm{d}x = 1.$$

The probability of an interval is the integral of the density over that interval. A density can exceed one because it has units; only integrated probabilities must lie between zero and one.

Variance and standard deviation are

$$\mathrm{Var}(X) = \langle X^2 \rangle - \langle X \rangle^2, \qquad \sigma_X = \sqrt{\mathrm{Var}(X)}.$$

Conditional probability and Bayes' theorem are

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}.$$

Events are independent only if $P(A \cap B) = P(A)P(B)$.

Table 2 summarizes the course distributions. Match the mechanism to the distribution: fixed independent trials give binomial counts; constant-rate independent events give Poisson counts and exponential waiting times.

A.12 Waiting times and survival probabilities

A waiting time T is the time until an event occurs, such as a molecule unbinding from a receptor. The probability density $p(t)$ describes how likely the event is to occur near time t. The survival probability $S(t)$ is the probability that the event has not yet occurred by time t:

$$S(t) = P(T > t).$$

As $S(t)$ decreases, the lost probability appears in the waiting-time density, so

$$p(t) = -\frac{\mathrm{d}S}{\mathrm{d}t}.$$

If the event occurs with a constant rate λ, the waiting time is exponentially distributed:

$$S(t) = e^{-\lambda t}, \qquad p(t) = \lambda e^{-\lambda t}, \qquad \langle T \rangle = \frac{1}{\lambda}.$$

The mean waiting time is therefore the inverse of the event rate.

Worked example

For $k_{\mathrm{off}} = 0.2\,\mathrm{s}^{-1}$, the mean bound lifetime is $5\,\mathrm{s}$ and $P(T > 10\,\mathrm{s}) = e^{-2} \approx 0.135$.

A.13 Combinatorics and binomial coefficients

The factorial $n! = n(n-1)\cdots 1$ counts orderings of n distinct objects. The number of unordered choices of k objects from n is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

For Ω sites and L indistinguishable ligands with at most one ligand per site,

$$W(L) = \binom{\Omega}{L}, \qquad \frac{W(L-1)}{W(L)} = \frac{L}{\Omega - L + 1} \approx \frac{L}{\Omega}$$

in the dilute limit.

For n independent trials with success probability p,

$$P(K = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

A.14 Boltzmann factors, partition functions, and free energy

At thermal equilibrium, a system can occupy states indexed by i, each with energy E_i. The Boltzmann factor

$$w_i = e^{-\beta E_i}, \qquad \beta = \frac{1}{k_B T},$$

is the relative statistical weight of state i; states with lower energy have larger weight. Here k_B is Boltzmann's constant and T is absolute temperature, so $1/\beta = k_B T$ is the thermal energy scale. The partition function is the sum of all weights,

$$Z = \sum_i w_i = \sum_i e^{-\beta E_i}.$$

It provides the normalization needed to convert the weight of state i into its probability:

$$p_i = \frac{w_i}{Z} = \frac{e^{-\beta E_i}}{Z}.$$

For macrostates with multiplicities W_A, W_B and energies E_A, E_B,

$$\frac{P_B}{P_A} = \frac{W_B}{W_A} e^{-\beta(E_B - E_A)}.$$

Low energy and high multiplicity both increase probability. With $S = k_B \ln W$ and $F = E - TS$, the total macrostate weight is $e^{-\beta F}$.

Worked example: one ligand and one receptor

With one bound state of energy ϵ_b and Ω unbound states of energy zero,

$$p_b = \frac{e^{-\beta\epsilon_b}}{\Omega + e^{-\beta\epsilon_b}} = \frac{1}{1 + e^{\beta \Delta F}},$$

where $\Delta F = \epsilon_b + k_B T \ln \Omega$. Binding energy competes with the multiplicity of unbound positions.

A.15 Entropy and constrained optimization with Lagrange multipliers

Let p_i be the probability of state i, with $p_i \geq 0$ and $\sum_i p_i = 1$. The dimensionless Shannon entropy is

$$S = -\sum_i p_i \ln p_i.$$

It is zero for certainty and maximal for a uniform distribution when normalization is the only constraint.

To maximize $f(\mathbf{x})$ subject to $g(\mathbf{x}) = c$, define

$$\mathcal{L} = f - \lambda(g - c)$$

and solve $\nabla \mathcal{L} = 0$ together with the constraint.

Maximizing entropy subject to normalization and mean energy gives

$$\mathcal{L} = -\sum_i p_i \ln p_i - \alpha \left(\sum_i p_i - 1 \right) - \beta \left(\sum_i p_i E_i - \langle E \rangle \right).$$

Differentiation gives $p_i \propto e^{-\beta E_i}$. The multiplier β becomes physical inverse temperature only when the constrained quantity is energy in the appropriate thermodynamic setting.

A.16 Least squares, likelihood, and optimization

For observations (x_i, y_i) and predictions $\hat{y}_i = f(x_i; \boldsymbol{\theta})$, the residual is $r_i = y_i - \hat{y}_i$. Ordinary least squares minimizes

$$E(\boldsymbol{\theta}) = \sum_{i=1}^{N} r_i^2.$$

Worked example

Fitting $y = ax + b$ to $(0, 1)$, $(1, 2)$, and $(2, 2)$ gives the normal equations

$$5a + 3b = 6, \qquad 3a + 3b = 5,$$

so $a = 1/2$ and $b = 7/6$.

If errors are independent Gaussian variables with common variance σ^2, the likelihood is

$$\mathcal{L} \propto \exp\left[-\frac{1}{2\sigma^2}\sum_i r_i^2\right].$$

Maximum likelihood is then equivalent to least squares. Unequal variances instead motivate weighted least squares, $E = \sum_i r_i^2/\sigma_i^2$.

In matrix form, $\mathbf{y} = X\boldsymbol{\theta} + \boldsymbol{\epsilon}$ and

$$X^{\mathsf{T}} X\hat{\boldsymbol{\theta}} = X^{\mathsf{T}}\mathbf{y}.$$

At the solution, the residual is orthogonal to every column of X.

Residual structure should be inspected. A curved pattern, changing variance, or correlated residuals indicates a failed assumption. Within a nested family of models, adding parameters cannot increase the minimized in-sample squared error, but it may worsen prediction on new data.

A.17 Covariance, correlation, and principal component analysis

Covariance measures whether two variables tend to fluctuate together. It is positive when they tend to lie above or below their means at the same time, negative when one tends to be above its mean while the other is below, and near zero when there is no linear co-fluctuation. It is defined by

$$\mathrm{Cov}(X, Y) = \langle (X - \langle X\rangle)(Y - \langle Y\rangle)\rangle\,.$$

Correlation rescales it to a dimensionless number:

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Zero covariance excludes a linear association but not every nonlinear dependence.

For centered vector data $\delta\mathbf{x} = \mathbf{x} - \langle\mathbf{x}\rangle$, the covariance matrix is

$$C = \left\langle \delta\mathbf{x}\, \delta\mathbf{x}^{\mathsf{T}} \right\rangle.$$

The variance along a unit direction $\mathbf{v}$ is $\mathbf{v}^{\mathsf{T}} C\mathbf{v}$. The direction of greatest variance is the eigenvector with largest eigenvalue. The explained-variance fraction of component i is

$$\frac{\lambda_i}{\sum_j \lambda_j}.$$

Worked example

For centered data lying on $x_2 = x_1$,

$$C = s^2 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

The direction $(1,1)^{\mathsf{T}}/\sqrt{2}$ has eigenvalue $2s^2$ and contains all variation. The perpendicular direction has eigenvalue zero.

PCA should normally use centered data. Standardization may prevent variables with large units from dominating, but it also erases physically meaningful scale differences. PCA finds directions of variance, not automatically causal variables or dynamical modes.

B Detailed derivations from the course

These longer calculations support the conceptual development in the main text. Their algebra, assumptions, and interpretations are retained from the day-by-day notes.

B.1 Discrete Fourier modes and the long-wavelength decay law

Consider one cosine mode,

$$P_n(x,t) = A_n \cos(q_n x),$$

with

$$q_n = \frac{2\pi n}{L}.$$

Substituting this into the random-walk update rule gives

$$\begin{aligned} P_n(x, t+\Delta t) = A_n \cos(q_n x) &+ k\Delta t A_n \cos\left(q_n(x-a)\right) \\ &+ k\Delta t A_n \cos\left(q_n(x+a)\right) - 2k\Delta t A_n \cos(q_n x). \end{aligned}$$

Using

$$\cos(\alpha+\beta) + \cos(\alpha-\beta) = 2\cos\alpha\cos\beta,$$

we obtain

$$\begin{aligned} P_n(x, t+\Delta t) &= A_n \cos(q_n x) + 2k\Delta t A_n \cos(q_n x)\cos(q_n a) \\ &\quad - 2k\Delta t A_n \cos(q_n x) \\ &= A_n \cos(q_n x)\left[1 + 2k\Delta t\left(\cos(q_n a) - 1\right)\right]. \end{aligned}$$

Since

$$P_n(x,t) = A_n \cos(q_n x),$$

we have

$$\boxed{P_n(x, t+\Delta t) = r_n P_n(x,t),}$$

where

$$\boxed{r_n = 1 - 2k\Delta t\left[1 - \cos(q_n a)\right].}$$

The spatial shape $\cos(q_n x)$ has not changed. Only its amplitude has been multiplied by the number r_n.

Therefore,

$$\boxed{\cos(q_n x) \text{ is an eigenfunction of the random-walk update operator.}}$$

The corresponding eigenvalue for one time step is r_n.

Physical meaning of the Fourier modes

Each value of n corresponds to a different spatial wavelength,

$$\lambda_n = \frac{2\pi}{q_n} = \frac{L}{n}.$$

Small n corresponds to long-wavelength, slowly varying structures. Large n corresponds to short-wavelength, rapidly varying structures. Diffusion suppresses the short-wavelength modes more rapidly than the long-wavelength modes. This is why an initially sharp distribution becomes smooth

with time.

Small-angle approximation and the diffusion constant

For wavelengths much larger than the lattice spacing,

$$q_n a \ll 1.$$

Using the Taylor expansion

$$\cos\theta = 1 - \frac{\theta^2}{2} + \mathcal{O}(\theta^4),$$

we obtain

$$\cos(q_n a) \approx 1 - \frac{q_n^2 a^2}{2}.$$

Therefore,

$$\begin{aligned} r_n &= 1 - 2k\Delta t\,[1 - \cos(q_n a)] \\ &\approx 1 - 2k\Delta t \left(\frac{q_n^2 a^2}{2}\right) \\ &= 1 - ka^2 q_n^2 \Delta t. \end{aligned}$$

Define the diffusion constant

$$\boxed{D = ka^2.}$$

Then

$$\boxed{r_n \approx 1 - Dq_n^2 \Delta t.}$$

Hence

$$P_n(x, t + \Delta t) = P_n(x, t)\left(1 - Dq_n^2 \Delta t\right).$$

Evolution over many time steps

After N time steps,

$$P_n(x, t + N\Delta t) = r_n^N P_n(x, t).$$

In the long-wavelength approximation,

$$r_n^N = \left(1 - Dq_n^2 \Delta t\right)^N.$$

Let the total elapsed time be

$$\tau = N\Delta t.$$

Therefore,

$$N = \frac{\tau}{\Delta t},$$

and

$$r_n^N = \left(1 - Dq_n^2 \Delta t\right)^{\tau/\Delta t}.$$

Using

$$\lim_{\epsilon \to 0}(1 + \alpha\epsilon)^{1/\epsilon} = e^{\alpha},$$

we obtain

$$\boxed{r_n^N \longrightarrow e^{-Dq_n^2 \tau}.}$$

Thus a Fourier mode evolves as

$$\boxed{P_n(x,t) = A_n e^{-Dq_n^2 t}\cos(q_n x).}$$

The relaxation time of mode n is therefore

$$\tau_n = \frac{1}{Dq_n^2}.$$

Since q_n increases with n, high-frequency modes decay much faster than low-frequency modes.

B.2 The continuum limit: symmetry, Taylor expansion, and assumptions

Start from the discrete random-walk update rule,

$$P(x,t+\Delta t) = P(x,t) + k\Delta t\, P(x-a,t) + k\Delta t\, P(x+a,t) - 2k\Delta t\, P(x,t).$$

Equivalently,

$$P(x,t+\Delta t) = P(x,t) + k\Delta t\left[P(x-a,t) + P(x+a,t) - 2P(x,t)\right].$$

Here k is the hopping rate in each direction and a is the lattice spacing.

We now assume that $P(x,t)$ varies smoothly over distances comparable to a. This allows us to Taylor-expand the shifted probability distributions around x. For the right-shifted position,

$$P(x+a,t) = P(x,t) + a\frac{\partial P}{\partial x} + \frac{a^2}{2}\frac{\partial^2 P}{\partial x^2} + \frac{a^3}{3!}\frac{\partial^3 P}{\partial x^3} + \cdots.$$

For the left-shifted position,

$$P(x-a,t) = P(x,t) - a\frac{\partial P}{\partial x} + \frac{a^2}{2}\frac{\partial^2 P}{\partial x^2} - \frac{a^3}{3!}\frac{\partial^3 P}{\partial x^3} + \cdots.$$

Adding the two expansions causes all odd-order derivatives to cancel,

$$P(x+a,t) + P(x-a,t) = 2P(x,t) + a^2\frac{\partial^2 P}{\partial x^2} + \mathcal{O}(a^4).$$

Therefore,

$$P(x-a,t) + P(x+a,t) - 2P(x,t) = a^2\frac{\partial^2 P}{\partial x^2} + \mathcal{O}(a^4).$$

Substituting this result into the update rule gives

$$P(x,t+\Delta t) = P(x,t) + k\Delta t\, a^2\frac{\partial^2 P}{\partial x^2} + \mathcal{O}(a^4).$$

Now Taylor-expand the left-hand side in time,

$$P(x,t+\Delta t) = P(x,t) + \Delta t\frac{\partial P}{\partial t} + \mathcal{O}(\Delta t^2).$$

Substituting this gives

$$P(x,t) + \Delta t\frac{\partial P}{\partial t} = P(x,t) + k\Delta t\, a^2\frac{\partial^2 P}{\partial x^2}.$$

Canceling $P(x,t)$ from both sides gives

$$\Delta t \frac{\partial P}{\partial t} = k\Delta t\, a^2 \frac{\partial^2 P}{\partial x^2}.$$

Canceling Δt gives

$$\frac{\partial P}{\partial t} = ka^2 \frac{\partial^2 P}{\partial x^2}.$$

Defining

$$\boxed{D = ka^2,}$$

we recover

$$\boxed{\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}.}$$

Thus the continuum diffusion equation is the long-wavelength, smooth-space limit of the discrete random walk.

Why the first derivative disappears

The cancellation of the first spatial derivative is a consequence of the symmetry of the random walk. The walker moves left and right with the same rate. The term proportional to $+\partial P/\partial x$ from one direction therefore cancels the term proportional to $-\partial P/\partial x$ from the other. This leaves the second derivative as the leading spatial contribution,

$$\frac{\partial^2 P}{\partial x^2}.$$

If the walker instead had a systematic preference for one direction, the first derivatives would not cancel completely. The continuum equation would then contain a drift term in addition to diffusion. The continuum-limit derivation assumes that:

- the walk is unbiased, so left and right hopping rates are equal;
- the hopping rate k is independent of position and time;
- only nearest-neighbor jumps of length a occur;
- the process is Markovian, so the next transition depends only on the current state;
- $P(x,t)$ varies smoothly on the length scale a, allowing the spatial Taylor expansion;
- Δt is sufficiently small that the temporal Taylor expansion is valid;
- higher-order terms in a and Δt can be neglected;
- the continuum limit is taken while the combination $ka^2 = D$ remains finite.

B.3 Solving diffusion by separation of variables

We now recover the Fourier modes directly from the PDE.

Start from

$$\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}.$$

Assume a separable solution

$$P(x,t) = X(x)T(t).$$

Substitution gives

$$X(x)\frac{\mathrm{d}T}{\mathrm{d}t} = DT(t)\frac{\mathrm{d}^2X}{\mathrm{d}x^2}.$$

Divide by DXT:

$$\frac{1}{DT}\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{X}\frac{\mathrm{d}^2X}{\mathrm{d}x^2}.$$

The left-hand side depends only on t, while the right-hand side depends only on x. They can therefore be equal for every x and t only if both are equal to the same constant. For diffusive modes, write this constant as $-q^2$:

$$\frac{1}{DT}\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{X}\frac{\mathrm{d}^2X}{\mathrm{d}x^2} = -q^2.$$

This gives two ordinary differential equations:

$$\frac{\mathrm{d}T}{\mathrm{d}t} = -Dq^2T,$$
$$\frac{\mathrm{d}^2X}{\mathrm{d}x^2} = -q^2X.$$

Their solutions are

$$T(t) = Ce^{-Dq^2t},$$

and

$$X(x) = A\cos(qx) + B\sin(qx).$$

Therefore,

$$P(x,t) = \left[A\cos(qx) + B\sin(qx)\right]e^{-Dq^2t}.$$

Periodic boundary conditions require

$$q = q_n = \frac{2\pi n}{L}.$$

Thus

$$\boxed{P_n(x,t) = \left[A_n\cos(q_nx) + B_n\sin(q_nx)\right]e^{-Dq_n^2t}.}$$

Why this solves a general initial condition

The usefulness of Fourier analysis is not merely that a cosine happens to solve the equation. A sufficiently well-behaved periodic initial probability distribution can be decomposed into Fourier modes,

$$P(x,0) = A_0 + \sum_{n=1}^{\infty}\left[A_n\cos(q_nx) + B_n\sin(q_nx)\right].$$

Because the diffusion equation is linear, each Fourier mode evolve independently:

$$P(x,t) = A_0 + \sum_{n=1}^{\infty}\left[A_n\cos(q_nx) + B_n\sin(q_nx)\right]e^{-Dq_n^2t}.$$

Thus the original PDE has effectively been converted into independent equations for the amplitudes of the Fourier modes. For each mode,

$$\frac{\mathrm{d}A_n}{\mathrm{d}t} = -Dq_n^2A_n,$$

which is just an ordinary differential equation. This is the central simplification:

PDE in real space $\longrightarrow$ independent ODEs for Fourier modes.

B.4 Straight-line least squares: deriving the normal equations

At the minimum of the error function,

$$\frac{\partial E}{\partial a} = 0$$

and

$$\frac{\partial E}{\partial b} = 0.$$

Differentiating with respect to a gives

$$\frac{\partial E}{\partial a} = \sum_{i=1}^{N} -2x_i \left[y_i - (ax_i + b) \right] = 0.$$

Therefore,

$$a \sum_{i=1}^{N} x_i^2 + b \sum_{i=1}^{N} x_i = \sum_{i=1}^{N} x_i y_i.$$

Differentiating with respect to b gives

$$\frac{\partial E}{\partial b} = \sum_{i=1}^{N} -2 \left[y_i - (ax_i + b) \right] = 0.$$

Therefore,

$$a \sum_{i=1}^{N} x_i + Nb = \sum_{i=1}^{N} y_i.$$

The fitting problem has now become a system of two linear equations for the two unknown parameters a and b.

B.4.1 Matrix representation and explicit fitting parameters

The two equations can be written as

$$\begin{pmatrix} \sum_i x_i^2 & \sum_i x_i \\ \sum_i x_i & N \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum_i x_i y_i \\ \sum_i y_i \end{pmatrix}.$$

A general system of linear equations has the form

$$A\mathbf{x} = \mathbf{c}.$$

Formally, if A is invertible,

$$\mathbf{x} = A^{-1}\mathbf{c}.$$

For the straight-line fit, define

$$\Delta = N \sum_i x_i^2 - \left(\sum_i x_i \right)^2.$$

The slope is

$$\boxed{a = \frac{N \sum_i x_i y_i - \left(\sum_i x_i \right) \left(\sum_i y_i \right)}{\Delta}.}$$

The intercept is

$$\boxed{b = \frac{\left(\sum_i x_i^2 \right) \left(\sum_i y_i \right) - \left(\sum_i x_i \right) \left(\sum_i x_i y_i \right)}{\Delta}.}$$

B.5 Maximum entropy with normalization and a mean constraint

Suppose that the states are labeled

$$i = 1, \ldots, n.$$

The probabilities must satisfy the normalization constraint

$$\sum_{i=1}^{n} p_i = 1.$$

As an example, suppose that the mean state is also constrained to be

$$\sum_{i=1}^{n} i p_i = 4.5.$$

We want to find the distribution $\{p_i\}$ that maximizes

$$S = -\sum_{i=1}^{n} p_i \ln p_i$$

subject to these constraints. Introduce Lagrange multipliers α and β and define

$$\mathcal{L} = -\sum_{i=1}^{n} p_i \ln p_i - \alpha \left(\sum_{i=1}^{n} p_i - 1 \right) - \beta \left(\sum_{i=1}^{n} i p_i - 4.5 \right).$$

The stationary condition for every p_i is

$$\frac{\partial \mathcal{L}}{\partial p_i} = 0.$$

This gives

$$-\ln p_i - 1 - \alpha - \beta i = 0.$$

Therefore,

$$p_i = e^{-1-\alpha-\beta i}.$$

Equivalently,

$$p_i \propto e^{-\beta i}.$$

The remaining constants are determined by the normalization and mean-value constraints.

C Readings and resources

Biology
- John Gerhart and Marc Kirschner, *Cells, Embryos, and Evolution.*
- Bernhard and Michael Grzimek, *Serengeti Shall Not Die.*
- Erwin Schrödinger, *What Is Life? The Physical Aspect of the Living Cell.*
- James D. Watson and Andrew Berry, *DNA: The Secret of Life.*
- Ron Milo and Rob Phillips, *Cell Biology by the Numbers.*
- Bruce Alberts et al., *Essential Cell Biology.*
- Sean B. Carroll, *The Serengeti Rules.*
- Mark Ptashne, *A Genetic Switch.*
- Horace Freeland Judson, *The Eighth Day of Creation.*
- Paul Nurse, *What Is Life?*
- Philip Ball, *How Life Works.*
- Ed Yong, *An Immense World.*
- iBiology videos.

Physics
- Albert Einstein and Leopold Infeld, *The Evolution of Physics.*
- George Gamow, *Biography of Physics.*
- I. Bernard Cohen, *The Birth of a New Physics.*
- BBC, *Forces of Nature with Brian Cox.*
- Carl Sagan, *Cosmos.*

Mathematics
- Steven Strogatz, *The Joy of x: A Guided Tour of Math, from One to Infinity.*
- Steven Strogatz, *Infinite Powers: How Calculus Reveals the Secrets of the Universe.*
- David Spiegelhalter, *The Art of Statistics: Learning from Data.*
- Leonard Mlodinow, *The Drunkard's Walk: How Randomness Rules Our Lives.*
- Daniel Kunin, *Seeing Theory: A Visual Introduction to Probability and Statistics.*
- Matilde Marcolli, *Lumen Naturae: Visions of the Abstract in Art and Mathematics*
- 3Blue1Brown videos.

Scientific practice and historical notebooks
- Charles Darwin's notebooks.
- Isaac Newton's "Waste Book".

Science, philosophy, and society
- Sean M. Carroll, *The Big Picture: On the Origins of Life, Meaning, and the Universe Itself.*
- Carl Sagan, *Billions and Billions: Thoughts on Life and Death at the Brink of the Millennium.*
- Christine Rosen, *The Extinction of Experience: Being Human in a Disembodied World.*
- Frederick Douglass, *Narrative of the Life of Frederick Douglass, an American Slave.*

D Daily flow of the course

This chronological index records the original order of topics. The reorganized notes develop each topic at its conceptual destination rather than reproducing a second chronological narrative.

D.1 Day 1

1. The scale of things (Section 1.1).
2. Gorter–Grendel experiment and the lipid bilayer (Section 1.2).
3. The cell as a bag of proteins (Section 2.2).
4. Biological turnover (Section 2.3).
5. mRNA production and degradation (Section 2.5).

D.2 Day 2

1. Theory of the mean (Section 2.1, 2.6).
2. Phytoplankton uptake and resource competition (Section 2.4).
3. Theory of the histogram (Section 3.3).
4. Biological turnover.

D.3 Day 3

1. One-dimensional symmetric random walk (Section 4.3, 4.8, 4.8.1, B.1).
2. From the random walk to the diffusion equation (Section 4.5).
3. Fick's law (Section 4.6).
4. Solving the diffusion equation by separation of variables (Section B.3).
5. Making a spike from cosines (Section 4.9).
6. Diffusion of an initially sharp distribution (Section 4.10).
7. Diffusion as a dynamical null hypothesis (Section 4.1, 4.2, 4.4, 4.7, 4.7.1, 4.8.2, B.2).
8. Representations and approximations (Section 6.2, 6.2.1, 6.3, A.3, A.4).
9. Null hypotheses and probability distributions (Section 3.1, 3.2, 3.2.1, Table 2, 3.4).

D.4 Day 4

1. The sameness of things (Section 7.1).
2. What is life? (Section 1, 7.2).
3. Binding by counting (Section 5.3).
4. Two receptors and interactions (Section 5.4).
5. Bicoid gradients (Section 5.5).
6. Allostery: the second secret of life (Section 5.6).

D.5 Day 5

1. Representations (Section 3.4, 6.1).
2. Fitting data (Section 6.3.1, 6.3.2, B.4, B.4.1).
3. Finding the right coordinates (Section 6.4, 6.4.1, 6.4.2, 6.4.3).
4. The eigenworm (Section 6.5).
5. Shannon entropy (Section 5.2, B.5).
6. Outlook: physical approaches to biology (Section 7.3).

E Notation summary

This table collects the recurring constants and notation conventions used throughout the notes. Variables specific to a single biological model are defined where that model is introduced.

Notation	Use in these notes
Δt, Δx	Finite increments in time or position.
$\mathrm{d}/\mathrm{d}t$	Ordinary derivative with respect to time, used when the quantity depends on a single independent variable.
$\partial/\partial t$, $\partial/\partial x$	Partial derivatives, used when a quantity depends on several variables.
$\mathcal{O}(\epsilon^n)$	Terms of order ϵ^n or smaller that are omitted from an approximation.
$\approx$, $\sim$, $\propto$	Approximately equal to, scales as, and proportional to, respectively.
$\langle X \rangle$	Mean or expectation value of the quantity X.
$\mathrm{Var}(X) = \langle X^2 \rangle - \langle X \rangle^2$	Variance of X.
$\mathrm{Cov}(X, Y)$	Covariance of X and Y.
μ, σ, σ^2	Mean, standard deviation, and variance of a probability distribution.
p_i	Probability assigned to the discrete state i.
$p(x)$	Probability density for a continuous variable x.
$\sum_i p_i = 1$, $\int p(x)\,\mathrm{d}x = 1$	Normalization conditions for discrete and continuous probability distributions.
$[X]$	Concentration of the molecular species X.
ss	Subscript denoting a steady-state value.
tot	Subscript denoting the total amount or concentration of a conserved quantity.
on, off	Subscripts denoting association and dissociation processes or the corresponding states and times.
$\mathbf{x}$, $\mathbf{p}$	Vectors; bold lowercase symbols denote collections of components.
A, C, X	Matrices or linear operators, as specified in the surrounding text.
$\hat{\boldsymbol{\theta}}$	Estimated value of a parameter vector.
$\delta\mathbf{x}$	Deviation of a vector from its mean, $\delta\mathbf{x} = \mathbf{x} - \langle \mathbf{x} \rangle$.
$\delta(x - x_0)$	Dirac delta distribution concentrated at $x = x_0$.
D	Diffusion constant.
J	Flux: the amount of material crossing a unit area per unit time.
$k_B = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$	Boltzmann constant.
T	Absolute temperature.
$\beta = 1/(k_B T)$	Inverse thermal energy.
E, S, F	Energy, entropy, and free energy.
Z	Partition function: the sum of the statistical weights of all allowed states.

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